

Action Plan for the Whitewater Watershed:
Claremont's Primary Source of Drinking Water



Prepared By:

May 2020

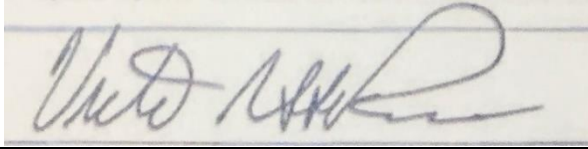
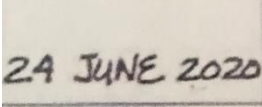
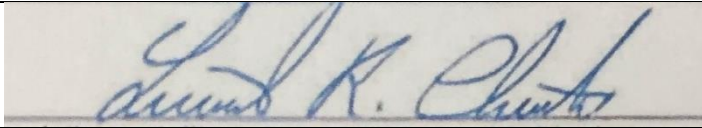
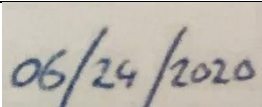
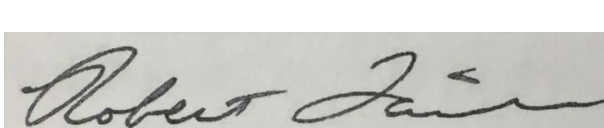
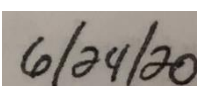
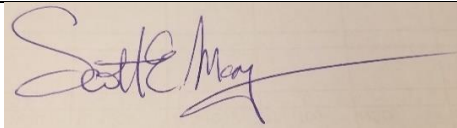
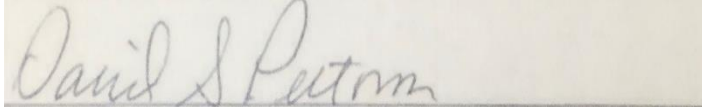
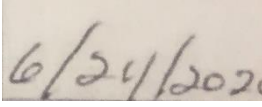
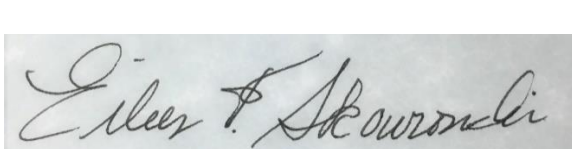
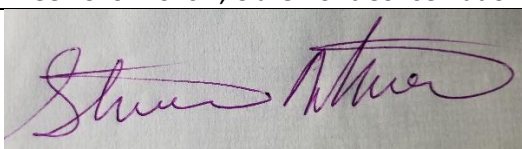
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Action Plan for the Whitewater Watershed,

Primary Source of Drinking Water for the City of Claremont's Public Drinking Water

As an ad hoc committee, we approve and endorse the Whitewater Watershed Action Plan and recommend its adoption by the Claremont Planning Board into the Master Plan and approval by the Claremont City Council.

		
Victor St. Pierre, Claremont Public Works Director		Date
		
Lionel Chute, Sullivan County Conservation District		Date
		6/24/2020
Corey Fitch, Cornish Conservation Commission		Date
		
Robert Lauricella, Utility Partners		Date
		6/24/2020
Scott Magnuson, Claremont Conservation Commission		Date
		
David Putnam, Claremont Planning Board		Date
		6/24/2020
Eileen Skowronski, Claremont Conservation Commission		Date
		6/24/2020
Steven Wilkie, Sullivan County ATV		Date

Thank you to the following individuals who provided assistance during the course of this project:

Meghan Butts, Upper Valley Lake Sunapee Regional Planning Commission

Keith Druhl, Executive Director, Claremont Community Television, Inc

Don Kretchmer, DK Water Resource Consulting LLC

Andrew Madison, NH Department of Environmental Services

Nancy Merrill, Director of Claremont Planning and Development

June 2020

Dear Reader,

Approximately 100 years ago, the City of Claremont purchased land in the Whitewater Watershed for the purpose of augmenting its public drinking water supply. The effort and foresight to establish this system was considerable. As we celebrate the 100th anniversary of the Whitewater Reservoir system, it is important that we evaluate its current conditions and look towards the future. The Reservoir supplies 70% of the City's water, is pristine in nature and is in all practical terms irreplaceable. It is one of the City's most valuable assets. To safeguard this source and look towards the future, the City of Claremont embarked on this sourcewater protection project.

With support from Granite State Rural Water Association and NH Department of Environmental Services, the City of Claremont initiated the formation of the Whitewater Watershed Advisory Committee in October 2019. This ad hoc committee is comprised of members from the three Watershed towns of Claremont, Cornish and Croydon. Members of the Committee have a diverse range of skills and perspectives. The Committee met on a monthly basis from November 2019 to March 2020 to evaluate current challenges and provide solutions. As a team we shared our knowledge of the region, identified potential issues and areas where additional information is needed and then developed action steps.

The timing of this initiative is right as the City contemplates updating its Forest Management Plan in the Watershed and we look to safeguarding this resource for another one hundred years. We have learned that land use and water quality are inextricably linked. The Committee recognizes that property owners in the Watershed and visitors alike have historically used this area for recreational purposes. Our recommendations are intended to balance the need for protection of water quality with other land uses within the Watershed.

This Action Plan is the result of the Committee's dedication and hours of work identifying issues, prioritizing risks, and developing a list of action items to promote water resource protection in the Watershed. We have made recommendations that are both regulatory and nonregulatory in nature with the firm belief that public education is paramount to success. The Whitewater Watershed Advisory Committee believes that the implementation of this plan will go a long way to preserve water quality in the Watershed. It has been a pleasure to work with Committee members and we especially thank members from neighboring towns for dedicating their time.

Respectfully Submitted,

Victor St. Pierre

Director, Claremont Public Works

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Executive Summary

The Whitewater Reservoir supplies more than 70% of the public drinking water for the City of Claremont. During the early part of last century, water from the Watershed was described as the most pristine in the State of New Hampshire (New Hampshire State Board of Health, 1916). Today this source is practically irreplaceable and should be safeguarded as such. Approximately 100 years ago, the City acquired this area for the purpose of developing a drinking water source. To celebrate the 100th anniversary, the City has undertaken a project to protect its primary source of drinking water.

This project began during the summer of 2019, when Granite State Rural Water Association (GSRWA) began to collate existing information about land ownership, zoning, and land conservation and develop associated maps. On October 1, 2019 a public meeting was held during which Victor St. Pierre, Claremont Public Works Director, Jennifer Palmiotto of Granite State Rural Water Association and Andrew Madison of NH Department of Environmental Services made presentations about Claremont's public drinking water system, source protection, and its primary source: the Whitewater Reservoir. Mr. St. Pierre and Dr. Palmiotto presented the same presentation to the Cornish Conservation Commission. In December 2019, Dr. Palmiotto made a similar presentation at the Croydon Selectmen's meeting. Stakeholders from the three Watershed communities of Cornish, Croydon and Claremont, were invited to participate on the Whitewater Watershed Advisory Committee. The resulting eight-member Committee had a variety of skills and perspectives ranging from members of town planning boards and conservation commissions, motorized recreational enthusiasts, to water supply professionals. The Committee met on a monthly basis from November 2019 through March 2020 to learn about Claremont's drinking water system by touring the treatment plant, reviewing potential risks to the Whitewater source and developing actionable steps to protect water resources in the Watershed.

Managing water resources at a watershed scale has been identified by the United States Environmental Protection Agency and the New Hampshire Department of Environmental Services as ecologically sound and practical. This is because watersheds and subwatershed units are recognized as the most practical management units for the development of local plans.

This Action Plan addresses the prevalent concerns of the Whitewater Watershed as identified by the Whitewater Watershed Advisory Committee. The Plan is divided into six chapters. Chapter One defines the term "watershed", describes the purpose of this Action Plan and the watershed approach, gives a brief history of watershed planning in the Whitewater Watershed, describes the collaborative nature of this watershed project, and lastly identifies how this plan can be used.

Chapter Two provides a physical description of the watershed including its size, characteristics of its ponds, streams and tributaries, wetlands, soils and geology. Land ownership and uses are

also described in this chapter. It should be noted that there is much to be learned about water quality in the Whitewater Reservoir due to the lack of raw water quality data.

Chapter Three provides information about the Claremont public drinking water system and a brief overview of its history. Potential Sources of Contamination to the Whitewater Reservoir are discussed, ranked and presented in a narrative form and also a table.

Chapter Four reviews existing regulatory and nonregulatory water resource protection measures in the Watershed. For example, land conservation, town master plans, and zoning ordinances are discussed.

Chapter Five presents the Committee's recommendations to promote water resources protection. Recommendations are presented in a narrative form and are also summarized in Table 5.1. These recommendations can be addressed through programs and projects developed by the Whitewater Watershed partners including: watershed municipalities, New Hampshire Department of Environmental Services, Colby Sawyer College, University of New Hampshire, New Hampshire Fish and Game, and the Whitewater Watershed Advisory Committee, among others.

Lastly, Chapter six provides conclusions about the plan and planning process.

Chapter 1. Introduction

1.1 Background

Although more than 70% of the Earth's surface is covered by oceans, lakes, rivers, and other bodies of water, only a small fraction of the 70% (2.4%) is fresh water. Of this small percentage of fresh water, nearly 90% is in the form of glaciers, ice caps, and snowfields. This means that only approximately 0.24% of the Earth's water is in fresh liquid form and available for human use. Preserving fresh water resources has long been recognized as a worthwhile goal. Fresh water is often vulnerable to both natural and human contamination, and therefore it is critical that this resource be managed wisely for the benefit of present and future generations.

In New Hampshire, there are approximately 60 surface sources (lakes, ponds, rivers, and reservoirs) which are used to supply water for public drinking water systems. These systems serve 40% of the State's population and the watersheds which drain to these sources span approximately 80% of New Hampshire (Model Rule for the Protection of Water Supply Watersheds, 2000). With this understanding of the limited nature of fresh water, this Action Plan aims to increase the understanding of water resources in the Whitewater Watershed, and to provide a meaningful foundation for decision-making. A watershed can be defined as a natural unit of land within which all water drains to a common outlet (Figure 1.1).

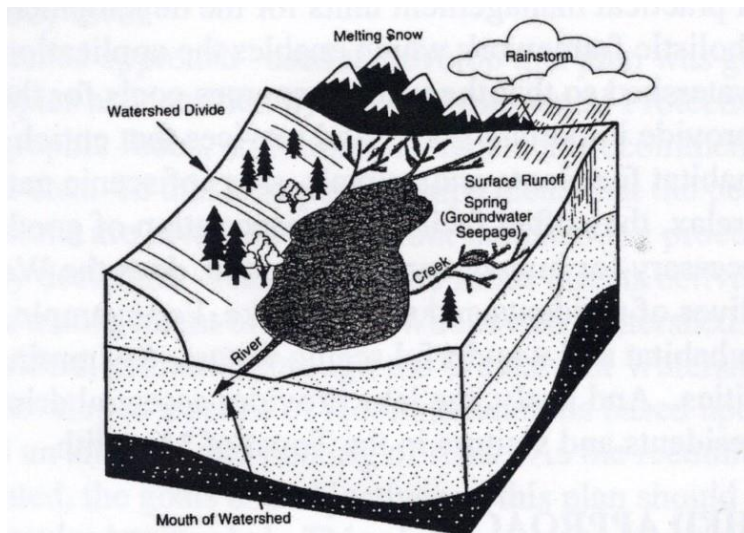
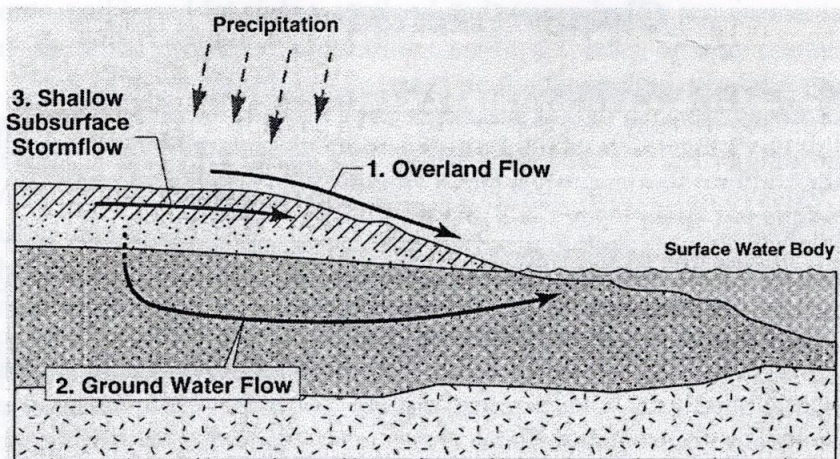


Figure 1.1 Depiction of a watershed.

A watershed includes two components: a surface water drainage basin and a groundwater drainage basin. The surface drainage basin is the land area from which all surface water flows drain toward a surface waterbody. The groundwater drainage basin is the land area and subsurface through which groundwater drains to a surface waterbody at a lower elevation



Source: Jeer et al, 1997.

Figure 1.2. The surface drainage basin may be larger or smaller than the groundwater drainage basin, depending on factors such as soils, slope, and surface cover.

One of the most important concepts is that surface water and groundwater are inextricably linked. For example, groundwater and surface water interact where groundwater discharges to lakes, rivers and in areas where ground conditions impede the drainage of water, such as in wetlands. This means that management of contamination and pollution sources throughout a watershed will benefit both groundwater and surface water. Watersheds may occupy tens to hundreds of square miles and cover several jurisdictions.

Managing water resources at a watershed scale has been identified as both ecologically sound and practical because watersheds and subwatershed units are recognized as the most practical management units for the development of local plans. A watershed plan is a holistic framework which enables the application of management tools within each subwatershed so that the water resources goals for the entire watershed are met.

Watersheds provide important goods and services that enrich our daily lives. They provide critical habitat for plants and animals, areas of scenic natural beauty, places to recreate and relax, they often facilitate transportation of goods and people, and provide fresh water necessary for human survival. The Whitewater Watershed enriches the lives of Claremont area residents and visitors alike. For example, the Whitewater Watershed contains the Whitewater Lower Reservoir, one of the City's most valuable and irreplaceable assets. The Reservoir provides 70% of the City's drinking water. Approximately 64% of the watershed is classified as "Highest Ranked Habitat in New Hampshire", and this region is prized for its recreational opportunities.

1.2 Purpose

The purpose of this project was to identify water resource protection opportunities and develop an action plan for their implementation. This project began during the summer of 2019, when Granite State Rural Water Association (GSRWA) started to collate existing information about land ownership, zoning, and land conservation and develop associated maps. On October 1, 2019 a public meeting was held during which Victor St. Pierre, Claremont Public Works Director, Jennifer Palmiotto of Granite State Rural Water Association and Andrew Madison of NH Department of Environmental Services presented information about Claremont's public drinking water system, source protection, and its primary source: the Whitewater Reservoir. Mr. St. Pierre and Dr. Palmiotto presented the same presentation to the Cornish Conservation Commission. In December 2019, Dr. Palmiotto made a similar presentation during the Croydon Selectmen's meeting. Stakeholders from the three Watershed communities of Cornish, Croydon and Claremont, were invited to participate on the Whitewater Watershed Advisory Committee. The resulting eight-member Committee had a variety of skills and perspectives ranging from members of town planning boards and conservation commissions, motorized recreational enthusiasts, to water supply professionals. The Committee met on a monthly basis from November 2019 through March 2020 to learn about Claremont's drinking water system by touring the treatment plant, reviewing potential risks to the Whitewater source and developing actionable steps to protect water resources in the Watershed.

1.3 The "Watershed Approach"

As early as the 1920's many federal agencies in the United States used watershed management for the purposes of controlling soil erosion and sedimentation. Increasingly, federal, state, and local agencies are focusing on non-point source pollution as a primary source of pollution to surface water and emphasizing the importance of planning at the watershed level. Watershed plans can work to improve water quality, manage recreational opportunities, maintain public health, or preserve the aesthetics of rivers and lakes. Community strategies for watershed planning have included the advent of partnerships and collaboration between the public, government agencies, and local organizations. Communities throughout the United States are increasingly coming to understand the importance of protecting watersheds in order to protect their water resources. As communities develop and the amount of watershed impervious cover increases in the form of parking lots, roads, and roof tops, the ability of a watershed to provide goods and services becomes impaired. This is because impervious surfaces collect and accumulate pollutants and when storm events occur, pollutants are more rapidly delivered to aquatic systems through runoff. Monitoring and modeling studies indicate that pollutant loads are directly related to watershed imperviousness. Research has shown that when impervious cover exceeds 10%, pollutant loads increase causing stream channels to become unstable and easily eroded (Schueler, 2002). When watershed imperviousness exceeds approximately 26%, streams become "non-supporting" meaning channel stability and biodiversity cannot be fully maintained even with the implementation of stormwater practices or retrofits (Schueler, 2002).

For these reasons, managing activities in a watershed is critical to its future well-being. Through use of the “watershed approach”, watershed associations, volunteer groups, government agencies and others can work together to protect ecosystem structure and function in order to safeguard water quality. In 1991 the United States Environmental Protection Agency defined the “watershed approach” as a coordinating framework for environmental management that focuses public and private sector efforts to address the highest priority problems within hydrologically-defined geographic area, taking into consideration both ground and surface water flow. According to New Hampshire Department of Environmental Services, “The watershed approach for management and planning is a strategy that has as its premise that many water quality and ecosystem problems are best solved at the watershed level rather than at the individual waterbody level.”

The “watershed approach” used to develop this plan was guided by three overarching principles as identified by the Environmental Protection Agency: partnerships, geographic focus, and sound management recommendations based on strong science and data. In this case partnerships mean that the people most affected by management decisions are involved throughout the planning process and are an integral part of shaping key decisions. The geographic focus directs activities and resources within the specific management unit of the Whitewater Watershed.

And lastly watershed stakeholders, the Whitewater Watershed Advisory Committee, developed a series of recommendations based upon available information and a ranking of risks. As the recommendations in this plan are implemented, the goals and objectives of this plan should be evaluated for effectiveness and revised as needed.

1.4 History of Planning and Water Quality Protection in the Whitewater Watershed”

Water Quality Protection goes back to the 1920’s when the City of Claremont purchased property in the Whitewater Watershed for the purpose of developing additional supply for the City’s water system. (See Chapter three for more history about the Claremont Water System).

In 1991 the Upper Valley Lake Sunapee Council prepared the Claremont Water Resources Management Protection Plan which identified threats to water resources, included an assessment of future water demand, and descriptions of 10 watersheds including the Whitewater/Redwater watershed.

In 2008 a Forest Management Plan was developed by Meadowsend Timberlands Ltd. For the City’s property around the Reservoir. While the focus of the plan was primarily on timber management, the plan identified two areas to be designated as “Ecological Reserve”. According to the management plan these reserve areas will have “no silvicultural management” and should be incorporated into the recreational and educational components as well as benefit the diversity of wildlife habitat.” “Unacceptable uses include motorized traffic including logging

equipment, infrastructure including benches or shelters.” (Meadowsend Timberlands Ltd., 2008). Although the concept of Ecological Reserves was identified in the plan, to date this concept has not been put into practice in any recognizable form on the ground.

The 2009 Cornish Master Plan specifically states that “Cornish should continue to respect this water source in its land use decisions” (Cornish Master Plan, 2009, Page 90). And further the plan emphasizes that a conservation zone should be created in the watershed. See Chapter 5 of this report for further discussion about the Cornish Master Plan.

In the City of Claremont’s 2017 Masterplan, water resource goals are highlighted in Chapter 7 and specifically refer to creating and adopting a watershed management plan to regulate land use activities that affect quality and quantity of the City’s surface and groundwater resources.

1.6 Use of this Plan

This Action Plan may be used to:

- Serve as a guidance document to assist the communities of Claremont, Cornish and Croydon in their planning efforts to protect water quality in the Whitewater Watershed.
- Guide New Hampshire Department of Environmental Services and other state and federal agencies in their efforts to protect and improve State surface waters.
- Outline the primary water quality and drinking water protection issues, based upon existing data and local knowledge.
- Identify top watershed management concerns and recommendations to address these concerns.
- Provide background and context on the Whitewater Watershed and its waterbased resources.
- Develop project ideas related to water quality or water resources improvements.
- Help identify technical or financial resources.
- Identify the technical or financial need of potential projects and partners.
- Support grant proposals.
- Provide guidance to local and regional planning and zoning processes.

It should be noted that this action plan represents the first step of a multi-stage process to protect the water resources in the Whitewater Watershed. As management activities are implemented and conditions change in the watershed, goals and objectives will need to be changed and the plan will need to be updated to reflect these changes. As watersheds are in a constant state of change, so too should plans reflect their ever-changing nature.

Chapter 2. Description of the Whitewater Watershed

2.1 Physical Description

The Whitewater Watershed is a subwatershed unit located within the western region of the HUC 12 0108010604 watershed (Figure 2.1). The Whitewater Watershed is defined as the area of land and the complex of wetlands, ponds, and tributaries which drain to the Whitewater Reservoir. The Watershed is a predominantly forested with very little residential development.

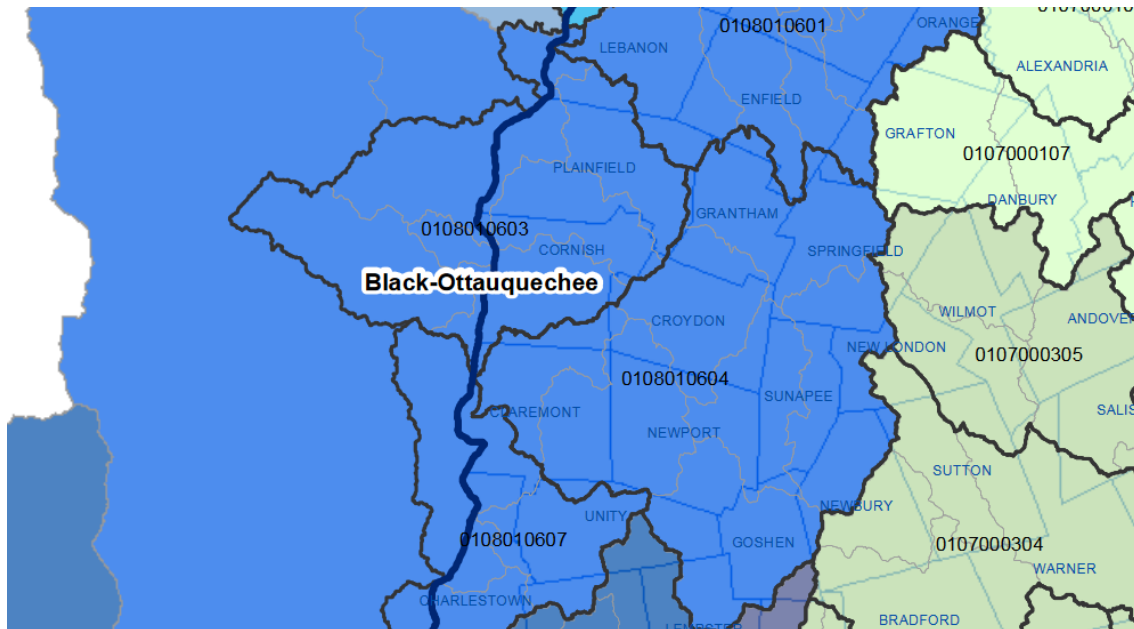


Figure 2.1 Map showing the Black-Ottauquechee watershed, HUC 12 0108010604. The Whitewater Watershed is a subwatershed located in the western region.

For the purposes of this plan the Watershed boundary was traced using the NRCES HUC 10 Sugar River watershed boundary. The Whitewater Watershed was further defined using 20' elevation contours acquired from USGS and in congruence with the Federal Standards and Procedures for the National Watershed Boundary Dataset.

The Watershed spans approximately 2,702 acres or 4.22 square miles and includes portions of the three towns of Claremont, Cornish and Croydon, New Hampshire. While the Whitewater Reservoir is the primary source of Claremont's drinking water, only 172 acres or 6% of the total Watershed lay within the City of Claremont. The majority of the Watershed, 1,803 acres or 67%, is located in the Town of Cornish. The remaining 27% or 728 acres is located in the Town of Croydon.

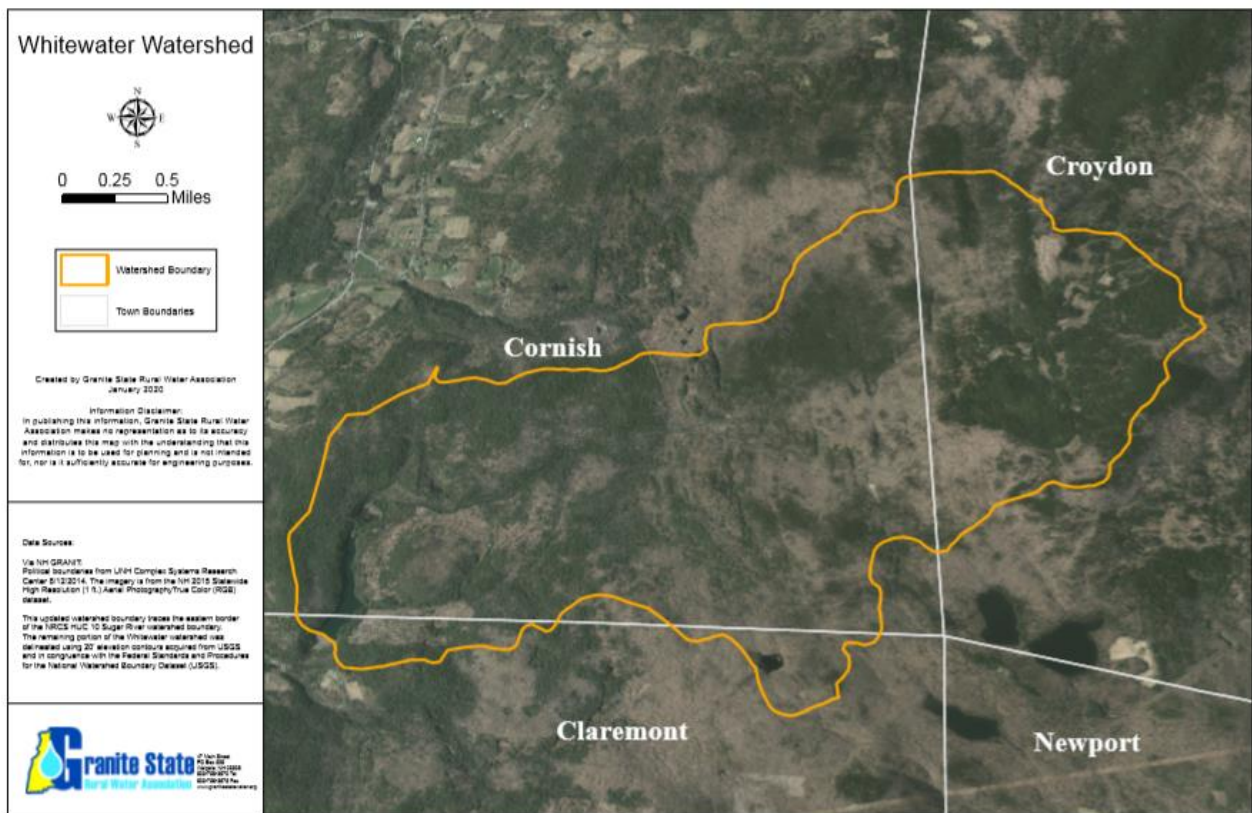


Figure 2.2 Map of the Whitewater Watershed spanning portions of the Municipalities of Claremont, Cornish and Croydon.

Table 2.1 Municipalities, associated acreage, and percent land cover in the Whitewater Watershed, New Hampshire.

Municipality	Area (Acres)	Percent of Watershed Area (%)
Claremont	172	6
Cornish	1,803	67
Croydon	728	27

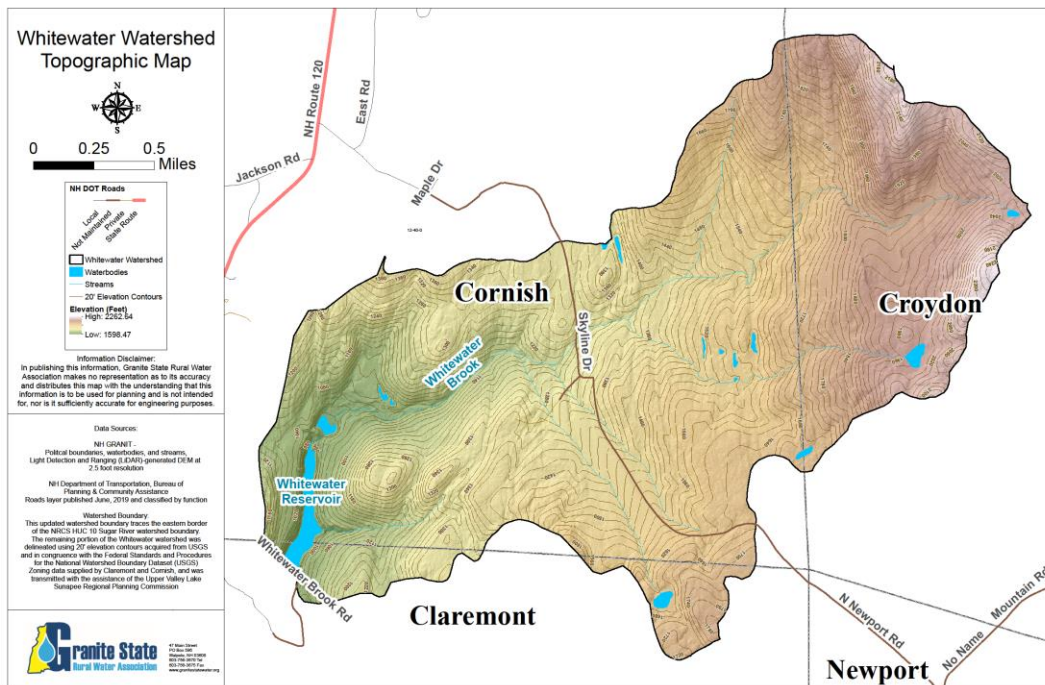


Figure 2.3 Topographic Map of the Whitewater Watershed

Within the Watershed topography ranges from 965 feet above sea level at the Reservoir to 2,240 in the Corbin Game Reserve in the northeastern portion.

Water Resources

The Watershed includes two impounded water bodies: the Whitewater Lower Reservoir and the formally breached Upper Reservoir. There are 13 additional ponds, 36.4 acres of wetlands and 9.2 miles of streams.

The Upper Reservoir was created in 1923 and further developed in 1943 when the dam was raised to a height of 62 feet. Its original storage capacity was approximately 420 acre feet or approximately 136 million gallons. A 1951 NH Fish and Game report indicated that the maximum depth sounded was 49 feet and the average depth was 25 feet. The reservoir was stocked with brook trout yearlings in 1948 and 1952. Rainbow trout yearlings were stocked in 1949, 1950, and 1951. Fishing was once permitted from shore. The dam for the Upper Reservoir failed in May, 1972 due to heavy rains. Floods caused a portion adjacent to the concrete spillway to give way. The breach created considerable erosion and damage (Singh, 2013).

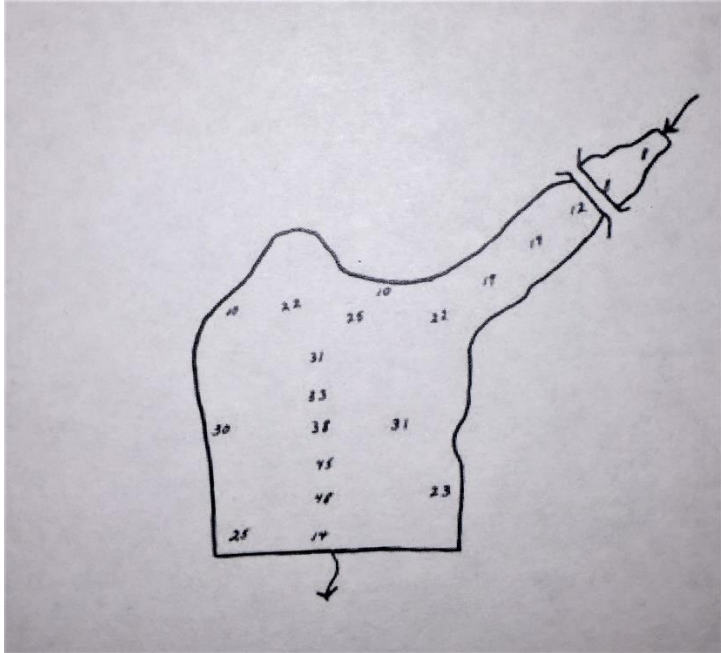


Figure 2.4 Schematic showing depth and shape of the Upper Reservoir prior to breaching in 1972. Source: 1977. NH Fish and Game report.

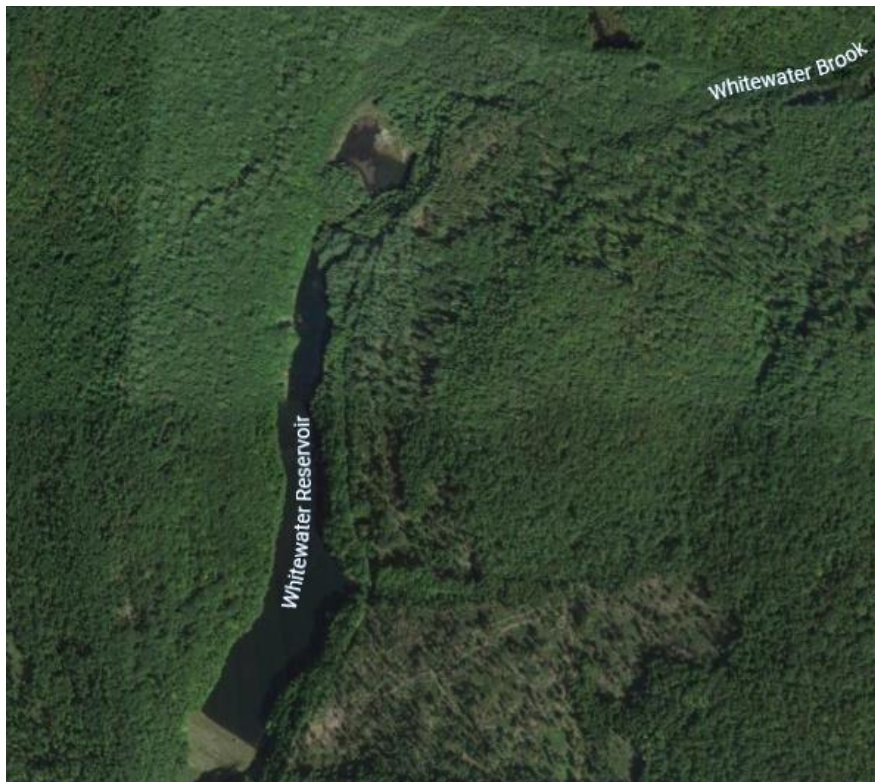


Figure 2.5. Aerial photo showing the Upper Reservoir with its broken dam and the Lower Reservoir below it. Source: Google Earth

The Lower Reservoir was created in 1965. This 17-acre impounded water body, runs approximately 3,884 feet in length and 663 feet at the widest point. The Reservoir has a length to width ratio of 5.85 and a total shoreline of 6,368 feet. When originally created, the Lower Reservoir reportedly had a storage capacity of approximately 150 million gallons.

A lake's shape controls much of what goes on physically and biologically underwater. The Reservoir's surface elevation (feet above sea level) is 965 feet (NH DES Consolidated List of Water Bodies, 2020). The Reservoir's greatest drop in water levels is reportedly twenty feet (2020 Pers. Comm. Chris Bergeron). Outflow is through the dam structure on the southern end.

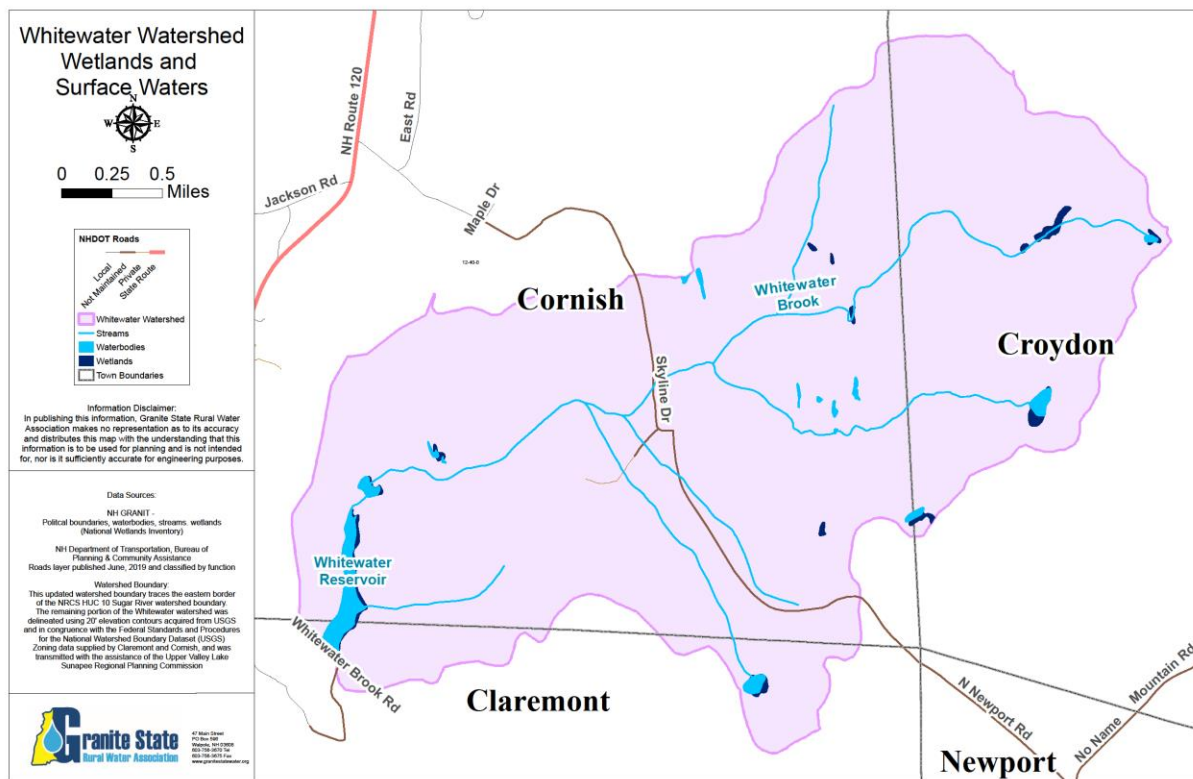


Figure 2.6 Map of Wetlands and surface waters in the Whitewater Watershed.

The Reservoir is fed by the outflow of what used to be the Upper Reservoir and two first order stream tributaries. Whitewater Brook originates approximately 2.5 miles to the east on the western flanks of the Croydon Mountains (Figure 2.4). The Brook and its tributaries pass through remote forest, and flow to a system of wetlands including forested wetland, shrub wetland and beaver meadows before draining into what was once the Upper Reservoir. From there the Brook drains into the Lower Reservoir. A second tributary enters the Reservoir on the eastern shore.

The Whitewater Brook dries up most summers during the months July-September unless it rains creating temporary stormwater runoff. In 2010 there was an algae bloom in the Reservoir. Copper sulfate was added to kill the algae. There has been no evidence of a bloom since that time.



Figure 2.7 At left: photo of Whitewater Brook as it enters the Upper Reservoir. At right: photo of a small unnamed tributary that drains directly into the eastern side of the Reservoir. Photo Credit: Meadowsend Timberlands, Ltd, 2008 Management Plan.

Results from various NHDES monitoring programs are summarized and reported as public documents every two years to the United States Environmental Protection Agency (USEPA) and the United States Congress. NH DES uses assessment units as the basic unit of record for conducting and reporting the results of water quality assessments. Assessment units are intended to be representative of homogenous units. Sometimes assessment units represent an entire water body. All of New Hampshire's water bodies have a fish consumption advisory or ban in effect for the general population for one or more fish species due to atmospheric deposition of because of mercury.

The 305(b)/303(d) Assessment Summary Reports for years 2008, 2010, 2012, 2014, 2016, and 2018 are available for the Whitewater Watershed, but provide limited data. Information from 2018 is presented below. Data from the 2018 report shows that all three sampling points were classified as "Full Support" for drinking water supply. The Whitewater Reservoir is classified as Class A water body as described by the Legislative Classification of Surface waters.

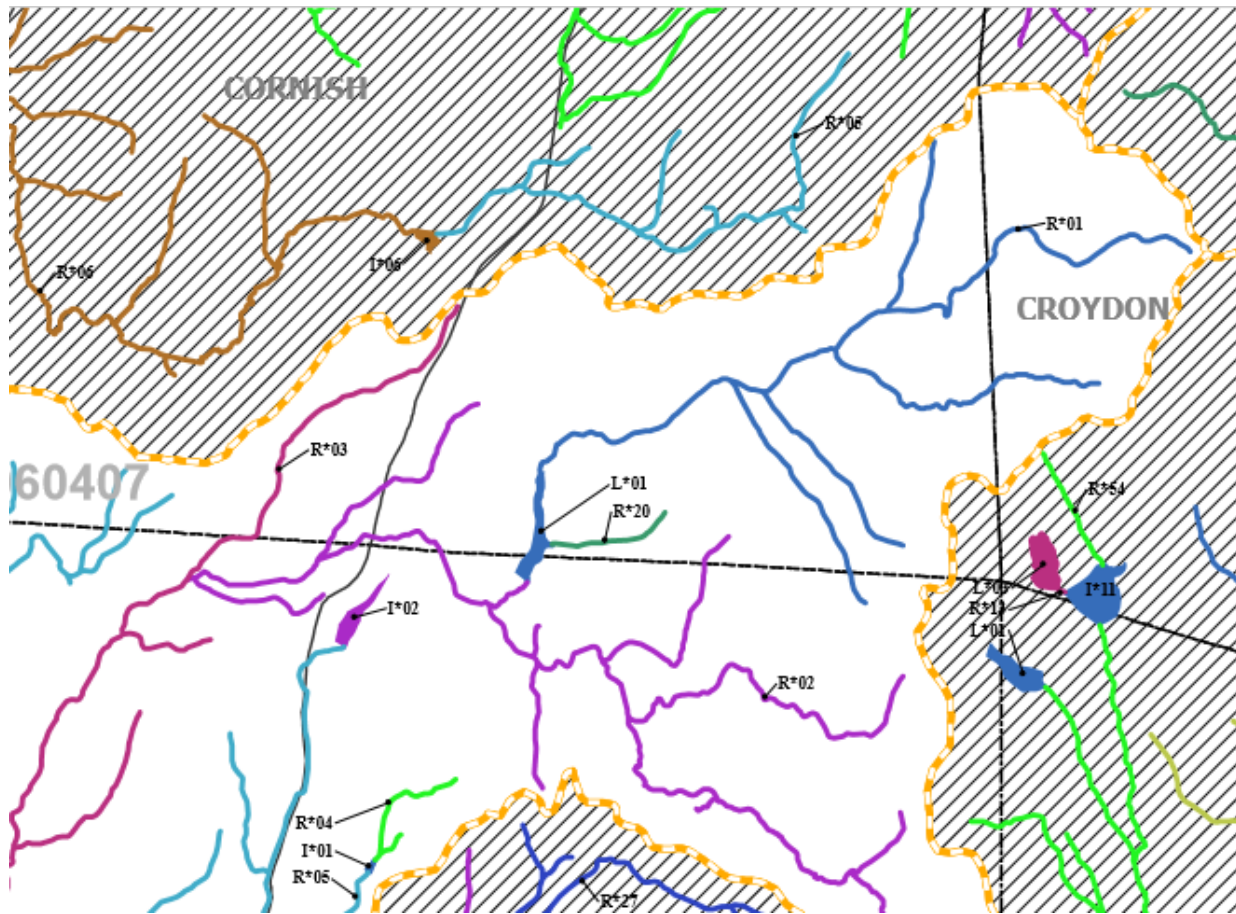


Figure 2.8 Map showing sampling points R*01, L*01, R*20 in the Whitewater Watershed from the 2018 NHDES Watershed 305(b)/303(d) Assessment Report.

Table 2.2 NH DES Watershed 305(b)/303(d) Assessment Report Information for the Whitewater Watershed.

Assessment Unit ID

NHLAK01060407-01

Assessment Unit Name

WHITEWATER RESERVOIR

Primary Town

CORNISH

Size

16.9080

ACRES

Beach

N

Assessment Unit Category*

3-ND

2018, 305 (b) /303 (d) -

All Reviewed Parameters by

Assessment Unit

Designated Use Description	*Desig. Use Category	Parameter Name	Parameter Threatened (Y/N)	Last Sample	Last Exceed	Parameter Category*	TMDL Priority
Aquatic Life Integrity	3-ND	Chlorophyll-a	N	N/A	NLV	3-ND	
		Dissolved oxygen saturation	N			3-ND	
		Oxygen, Dissolved	N			3-ND	
		pH	N			3-ND	
Fish Consumption	4A-M	Mercury	N			4A-M	
Potential Drinking Water Supply	2-G						
Primary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Secondary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Wildlife	3-ND						

Assessment Unit ID	NHRIV801060407-01	Size	7.9890 MILES	2018, 305(b)/303(d) -
Assessment Unit Name	WHITENATER BROOK - UNNAMED BROOKS	Beach	N	All Reviewed Parameters by
Primary Town	CORNISH	Assessment Unit Category*	3-ND	Assessment Unit

Designated Use Description	*Design. Use Category	Parameter Name	Parameter Threatened (Y/N)	Last Sample	Last Exceed	Parameter Category*	TMDL Priority
Aquatic Life Integrity	3-ND	Benthic-Macroinvertebrate Bioassessments (Streams)	N			3-ND	
		Dissolved oxygen saturation	N			3-ND	
		Fishes Bioassessments (Streams)	N			3-ND	
		Oxygen, Dissolved	N			3-ND	
		pH	N			3-ND	
Fish Consumption	4A-M	Mercury	N			4A-M	
Potential Drinking Water Supply	2-G						
Primary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Secondary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Wildlife	3-ND						

Assessment Unit ID	NHRIV801060407-20	Size	0.5600 MILES	2018, 305(b)/303(d) -
Assessment Unit Name	UNNAMED BROOK	Beach	N	All Reviewed Parameters by
Primary Town	CORNISH	Assessment Unit Category*	3-ND	Assessment Unit

Designated Use Description	*Design. Use Category	Parameter Name	Parameter Threatened (Y/N)	Last Sample	Last Exceed	Parameter Category*	TMDL Priority
Aquatic Life Integrity	3-ND	Benthic-Macroinvertebrate Bioassessments (Streams)	N			3-ND	
		Dissolved oxygen saturation	N			3-ND	
		Fishes Bioassessments (Streams)	N			3-ND	
		Oxygen, Dissolved	N			3-ND	
		pH	N			3-ND	
Fish Consumption	4A-M	Mercury	N			4A-M	
Potential Drinking Water Supply	2-G						
Primary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Secondary Contact Recreation	3-ND	Escherichia coli	N			3-ND	
Wildlife	3-ND						

How are assessments coded in the report card?

Assessment outcomes are displayed on a color scale as well as an alpha numeric scale that provides additional distinctions for the designated use and parameter level assessments as outlined in the table below.

	Severe	Poor	Likely Bad	No Data	Likely Good	Marginal	Good
	Not Supporting, Severe	Not Supporting, Marginal	Insufficient Information – Potentially Not Supporting	No Data	Insufficient Information – Potentially Full Supporting	Full Support, Marginal	Full Support, Good
CATEGORY	Description						
Category 2	Meets standards					2-M or 2-OBS	2-G
Category 3	Insufficient Information						
Category 4	Does not Meet Standards;						
4A	TMDL* Completed	4A-P	4A-M or 4A-T				
4B	Other enforceable measure will correct the issue.	4B-P	4B-M or 4B-T				
4C	Non-pollutant (i.e. exotic weeds)	4C-P	4C-M				
Category 5	TMDL* Needed	5-P	5-M or 5-T				

* TMDL stands for Total Maximum Daily Load studies (<http://des.nh.gov/organization/divisions/water/wmb/tmdl/index.htm>)

January 3, 2020

Habitat

The entire watershed is classified as the VT-NH Upland eco-region. There are two predominant habitat types: hemlock-hardwood pine and northern hardwood conifer.

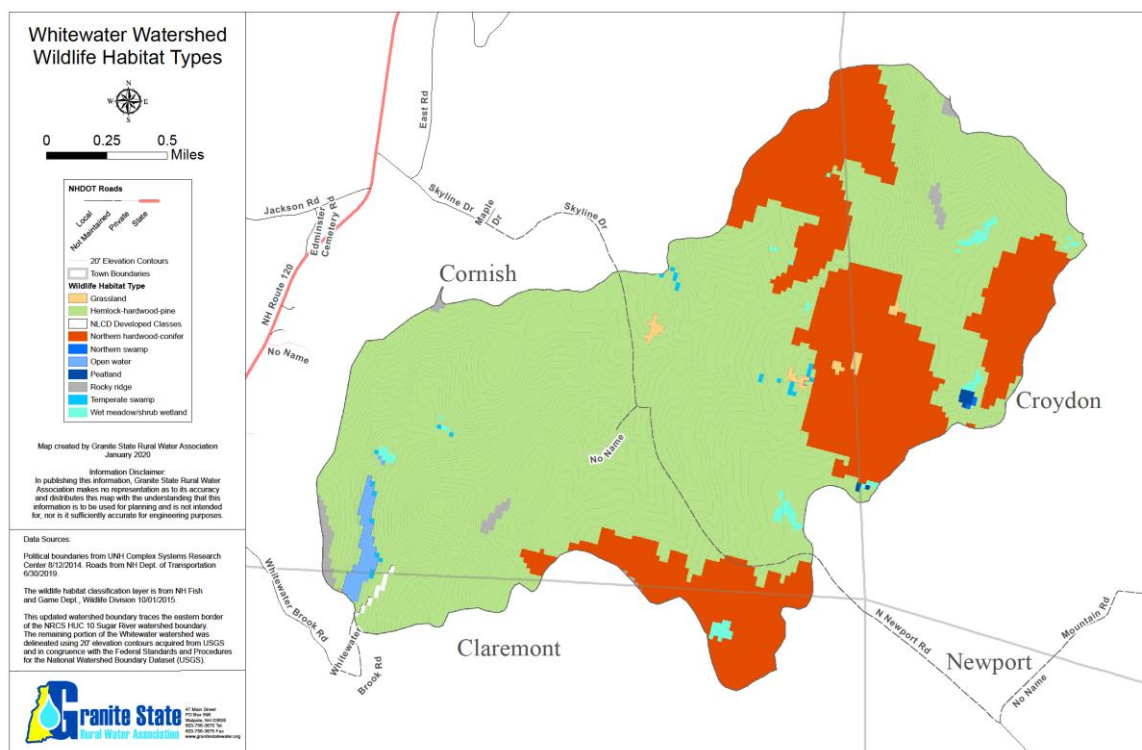


Figure 2.9 Wild life habitat types in the Whitewater Watershed.

The vastness of this unfragmented forest block makes this area significant for wildlife (Cornish NRI, 2013). This forested area contains numerous streams, seeps, vernal pools and rocky ledges. It provides excellent habitat for small and large mammals including moose, bear, bobcat, fisher, snowshoe rabbit, beaver, otter, fox, and coyote. According to the Cornish NRI, the Corbin Park fence retains the largest animals in the park, however wild mammals can typically cross this boundary by climbing trees, flying, crawling through the fence or through many large culverts. Approximately 1,734 acres or 64% of the watershed is classified as “Highest Ranked Habitat in NH” also known as Tier 1 wildlife habitat. Tier 1 wildlife habitat is of greatest habitat conservation priority because it represents the top 10-15% of habitat in the entire state. Approximately 521.5 acres or 19% of the watershed is ranked “Highest Ranked Habitat in Biological Region” (Fish & Game State, 2010).

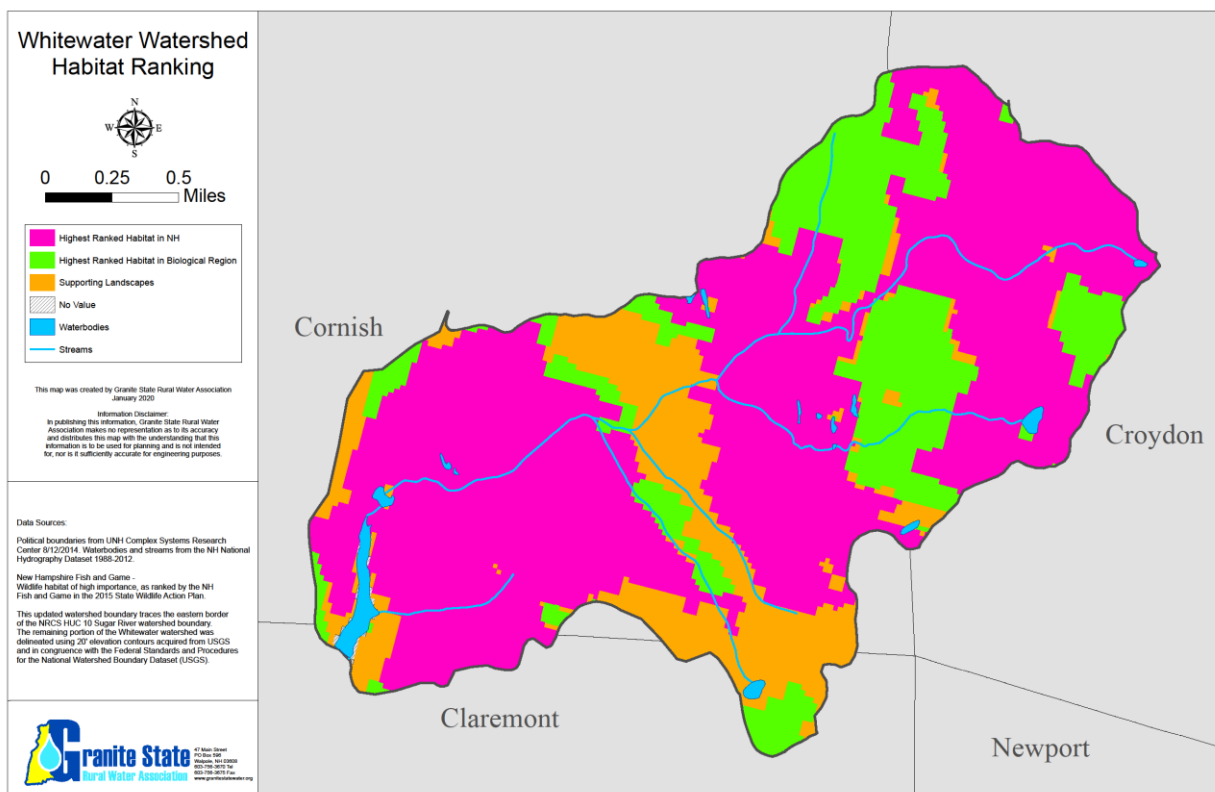


Figure 2.10 Map Showing Ranking of Habitat in the Whitewater Watershed according to the 2010 State Wildlife Action Plan

Soils and Geology

More than 85% of the Watershed is underlain by Spodosols. Spodosols are one of the eleven soil major orders in the world. These soils occur in climates where large amounts of water infiltrate the soil at one or more times per year. Spodosols commonly form in sandy parent materials under coniferous forest vegetation. As a consequence of their coarse texture, they have a high leaching potential. Spodosols are characterized by high acidity, and have a subsoil accumulation of organic matter, along with aluminum and iron oxides, called a spodic horizon. Typically low in natural fertility (basic cations, Ca^{2+} , Mg^{2+} , and K^{+}) and high in soil acidity (H^{+} , Al^{3+}), these soils require extensive inputs of lime and fertilizers to be agriculturally productive.

Within the Watershed there are areas comprising Histosols. Histosol soils are defined as having 16 inches or more of organic soil material in the upper 31 inches. Organic soil material has an organic carbon content of 12 to 18 percent by weight. Typically, Histosols have very low bulk density and are poorly drained because the organic matter has good water holding capacity. Most are Histosols are acidic and many are very deficient in major plant nutrients.

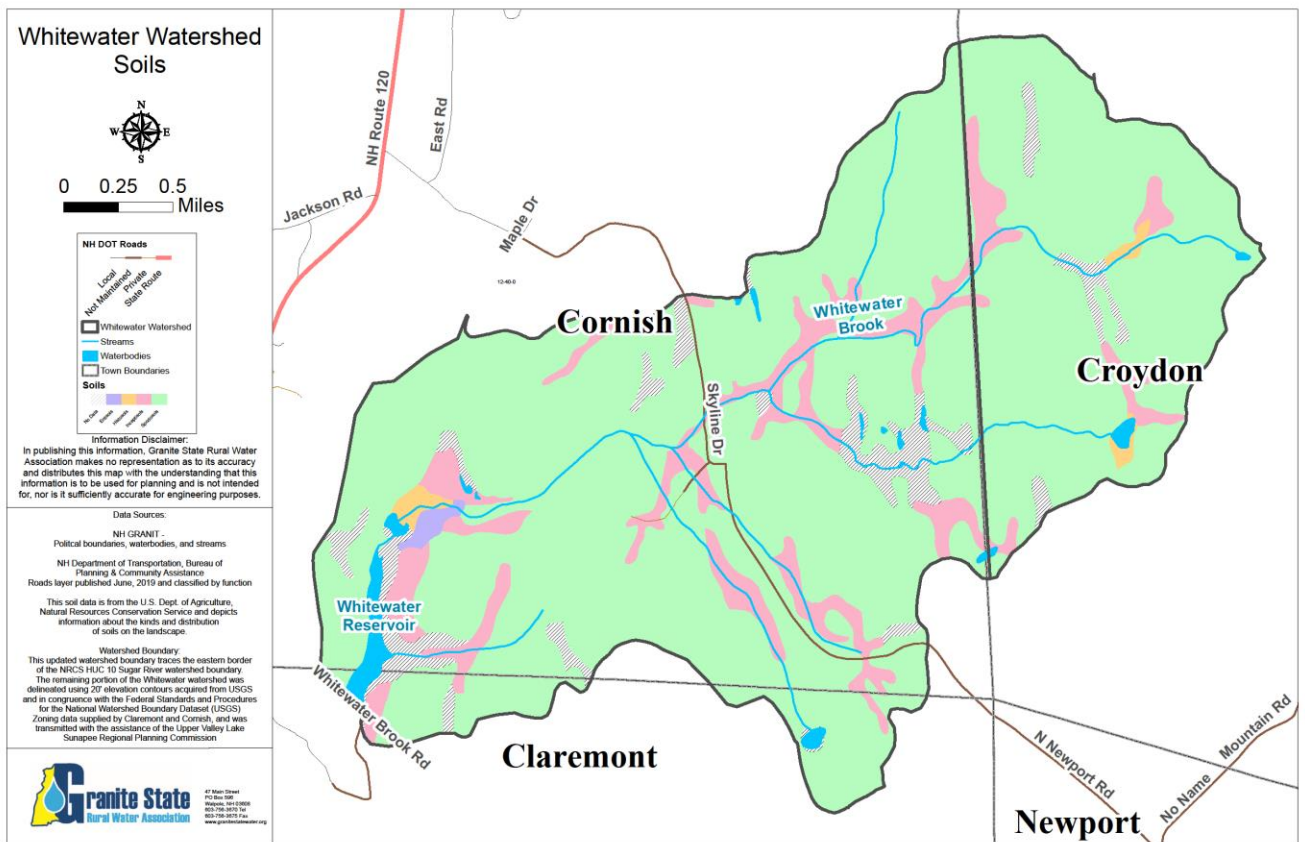


Figure 2.11 Map of Soil Orders in the Whitewater Watershed.

In the Watershed, the third soil order of note, is Inceptisols. These soils are relatively newly formed and have weak development of soil horizons. They have limited accumulation of clays, iron oxide, aluminium oxide or organic matter. They are the most abundant soil order on Earth.

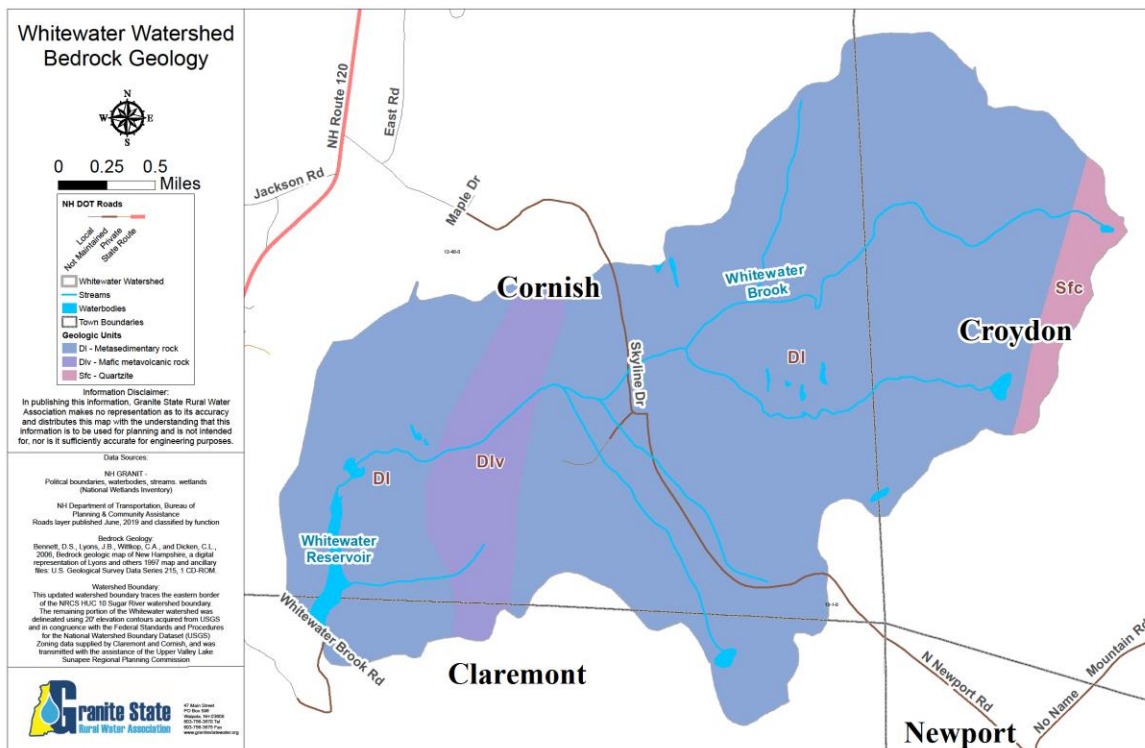


Figure 2.12 Map of bedrock geology in the Whitewater Watershed.

The bedrock geology of the Watershed is dominated by metasedimentary rock (Figure 2.11). This rock is first formed through the deposition and solidification of sediments. Next, the rock is buried underneath subsequent rock and subjected to high pressures and temperatures, causing the rock to recrystallize.

2.2 Land Use

Land Ownership

There are approximately 50 parcels that intersect or fall entirely within the Whitewater Watershed. Parcel data was reviewed from NH GRANIT, NH Parcel Mosaic data layer (February 2017) and updated with information from local tax maps from the Watershed municipalities. The accuracy of the resulting parcel map (Figure 2.13) is not survey quality, however, it identifies watershed parcels and property owners. A full list of property owners is located in Appendix D.

The City of Claremont owns five parcels within the Watershed. Three lots (8-1,8-2, and 18-2) are located in the City of Claremont. Two parcels (lot 12-43 and 12-38) are located in Cornish. All parcels except for 12-38 are adjacent to the Reservoir. Total City-owned acreage is 471 acres with 420 acres or 15.5% falling within Watershed boundaries.

Blue Mountain Forest Association is the largest property owner with 62% of the land or 1,674 acres in the watershed. The Association owns lots 13-1 and 14-1 in Cornish and 721 acres in the Watershed in Croydon. The Association manages this property as a private game reserve called Corbin Park.

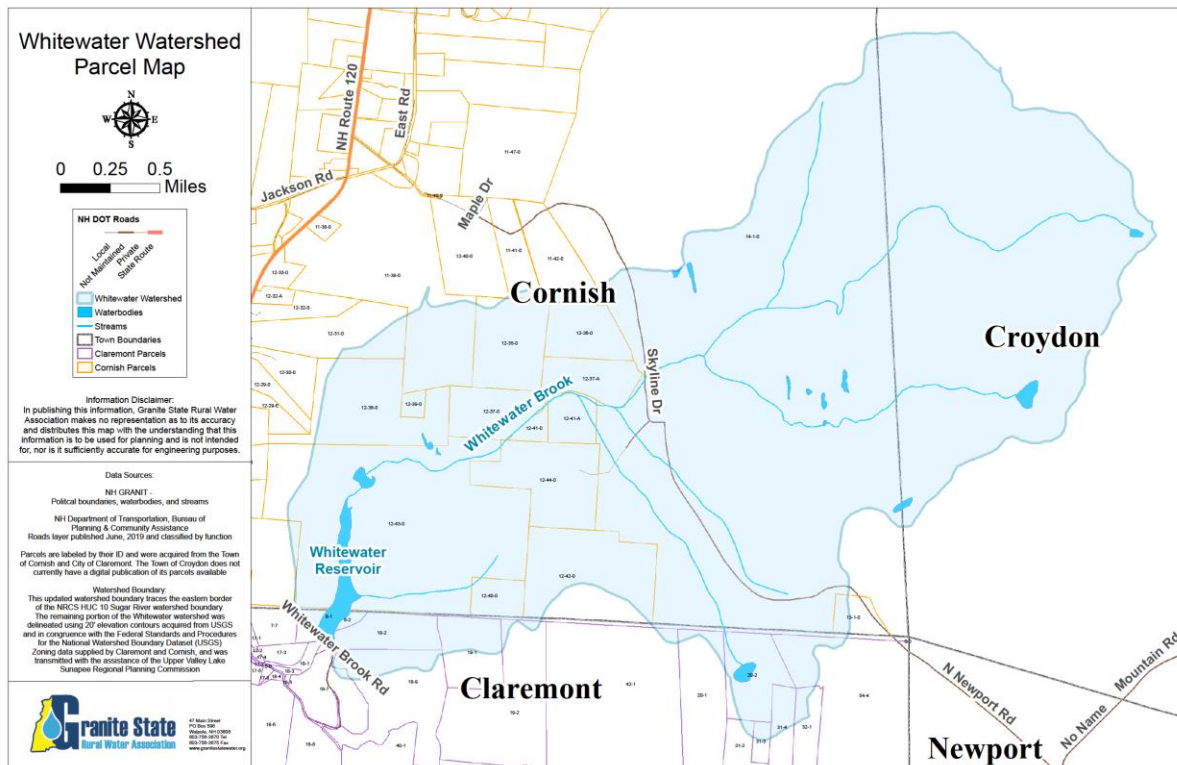


Figure 2.13 Map of parcels in the Whitewater Watershed.

Residential development is less than 15% in the Watershed. In the Claremont portion, there are two residential properties.

Recreational Use in the Watershed

Within the Whitewater Watershed there is a network of trails that is used by motorized recreational enthusiasts who enjoy snowmobiling, driving ATV's and 4x4s through the watershed and beyond. Trails include Skyline Drive, Cornish Turnpike and the Whitewater Trail.

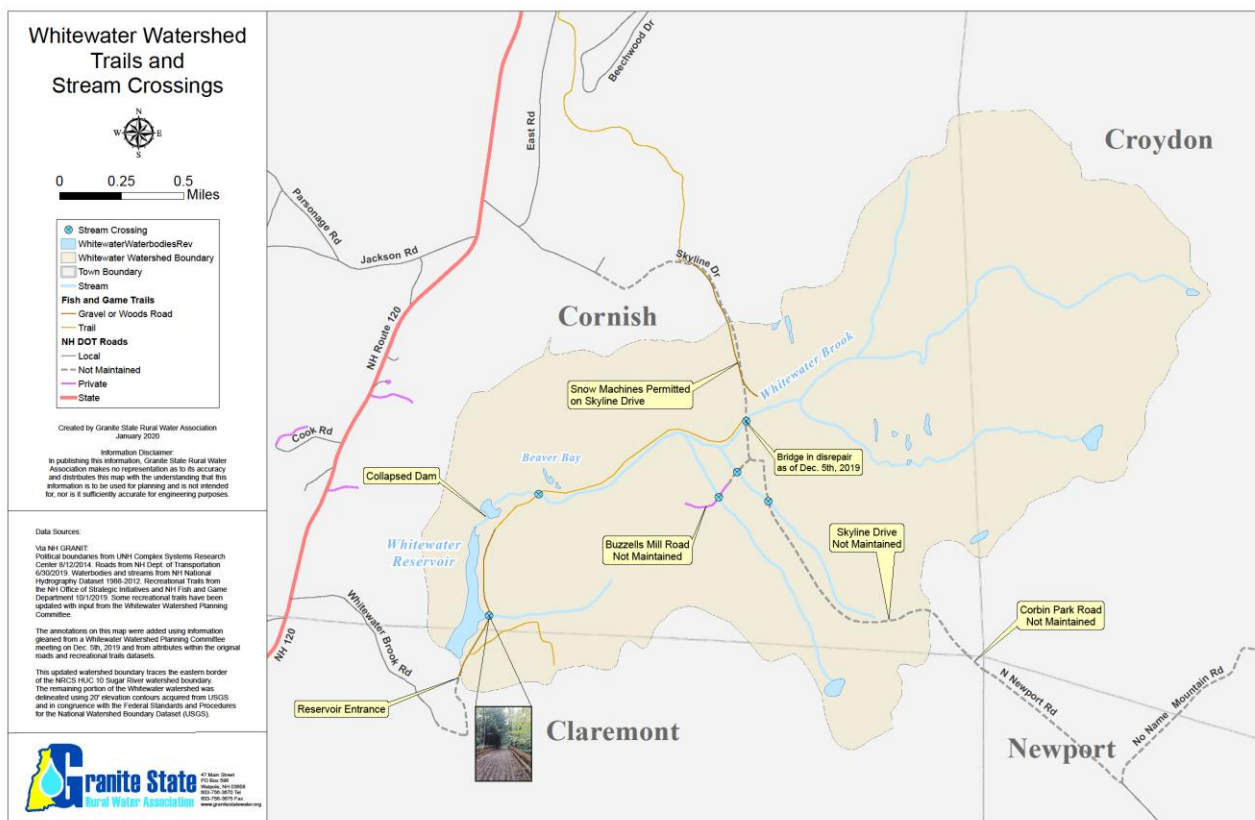


Figure 2.14 Map of trail network and stream crossings in the Whitewater Watershed.

In addition to motorized recreation, much of the Watershed is managed as a private game reserve. Blue Mountain Forest Association, the largest land owner in the Watershed, is the owner of Corbin Park. The property was established in 1891 and is New Hampshire's only private hunting preserve. The roughly 25,000-acre reserve spans portions of the towns of Newport, Cornish, Croydon, Grantham and Plainfield. The reserve is surrounded by a 26-mile-long fence, enclosing game animals such as elk, and boar creating a "cold weather climate safari atmosphere."

The park was originally created by Austin Corbin, a 19th-century railroad tycoon and banker who grew up in Newport, NH. After Austin Corbin died, Austin Corbin Jr.'s fortune decreased. When Austin Corbin Jr. died in 1938, he was practically penniless. That same year, the 1938 hurricane blew down large amounts of fencing allowing boar and elk to escape in large

numbers. The park fell into disrepair, until when in 1944 the Corbin family sold the park to a group of wealthy hunters. (Brown, 2016)

Today, details about the park are scarce and its employees are required to sign a nondisclosure agreement. The Association's membership is said to be about 30, limited by design and the five-figure cost of joining.

Timber Management

It is estimated that the Whitewater Watershed is more than 80% forested. Intact forest has a positive effect on water quality. In 2005, approximately 21,315 acres of land in the Town of Cornish were in the Current Use program. A large percentage of this acreage is within the Corbin Game Park Reserve, located within the Watershed (2009, Cornish Master Plan, Page 33). All of Corbin Park is managed as a registered tree farm.

In 2008 a forest management plan was developed for Claremont parcels 12-38 and 12-43. The plan describes the forest as a mix of white pine, hemlock and hardwoods. It varies from dry, south-facing -oak dominated sites, to a mix of pine, hemlock and hardwood on the mid-slopes, and hemlock-dominated areas everywhere else. The terrain is variable, but generally includes moderate slope with some rocky ground.

Timber harvest have taken place on lot 12-43 in 2010-2011 and 2012-2013. Harvested stands were hemlock-beech-oak-pine. Reportedly the logger was incentivized by the volume of timber cut as opposed to following the plan and harvest inspection was limited. Forest stands adjacent to Whitewater Brook are slated as the next area scheduled for harvest. The City anticipates developing a new forest management plan in the next 1-2 years.

The Sullivan County soil survey classifies soil types by their capability to support sufficient tree growth for commercial forestry operations. These are broken into five classes:

- Group 1A soils are the best for hardwood production because they are relatively deep, fertile and well drained.
- Group 1B are slightly less fertile and sandier than Group 1A so tree growth is not as vigorous.
- Group 1C is composed of outwash sands and gravels and are ideally suited to softwood production
- Group 2A soils are similar in productivity to Group 1A and Group 1B, but have physical features that make management more difficult such as steep slopes, erosion prone soils, rocky outcrops, surface boulders or extreme rockiness.
- Group 2B soils are poorly drained and therefore have lower productivity and significant management limitations.

The Watershed is dominated by Group 2A soils, soils which have management limitations.

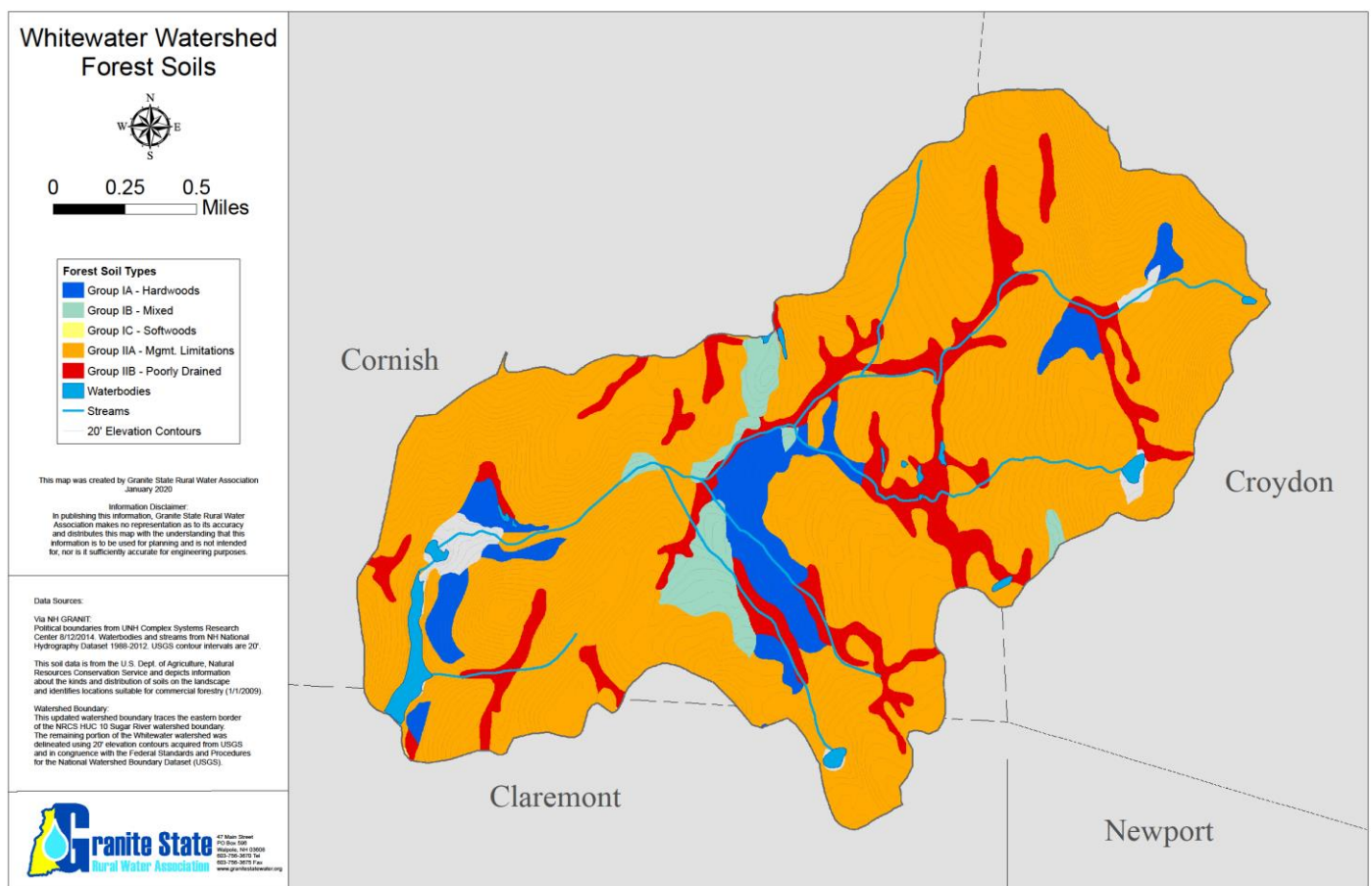


Figure 2.16 Map of Forest soils in the Whitewater Watershed.

Chapter 3. Claremont Water Supply

3.1 Description of the Claremont Water System

The Claremont water system is a public drinking water system, “a system for the provision to the public of piped water for human consumption if such system has at least 15 service connections or regularly serves an average of at least twenty-five individuals daily at least 60 days out of the year” (Chapter Env-ws 300 NH Drinking Water Rules). The Claremont public drinking water system serves approximately 9,000 people and has 3,600 service connections. Approximately 1.1 million gallons per day are treated and delivered into the distribution system. The Department has 5 full-time employees who maintain 90 miles of water distribution mains, 1,100 main line valves, 2 water storage tanks, 2 water pump stations, 435 hydrants, and more than 3,600 water meters. Daily operations of the water treatment plant are overseen by a Utility Partners, a contract operations company.

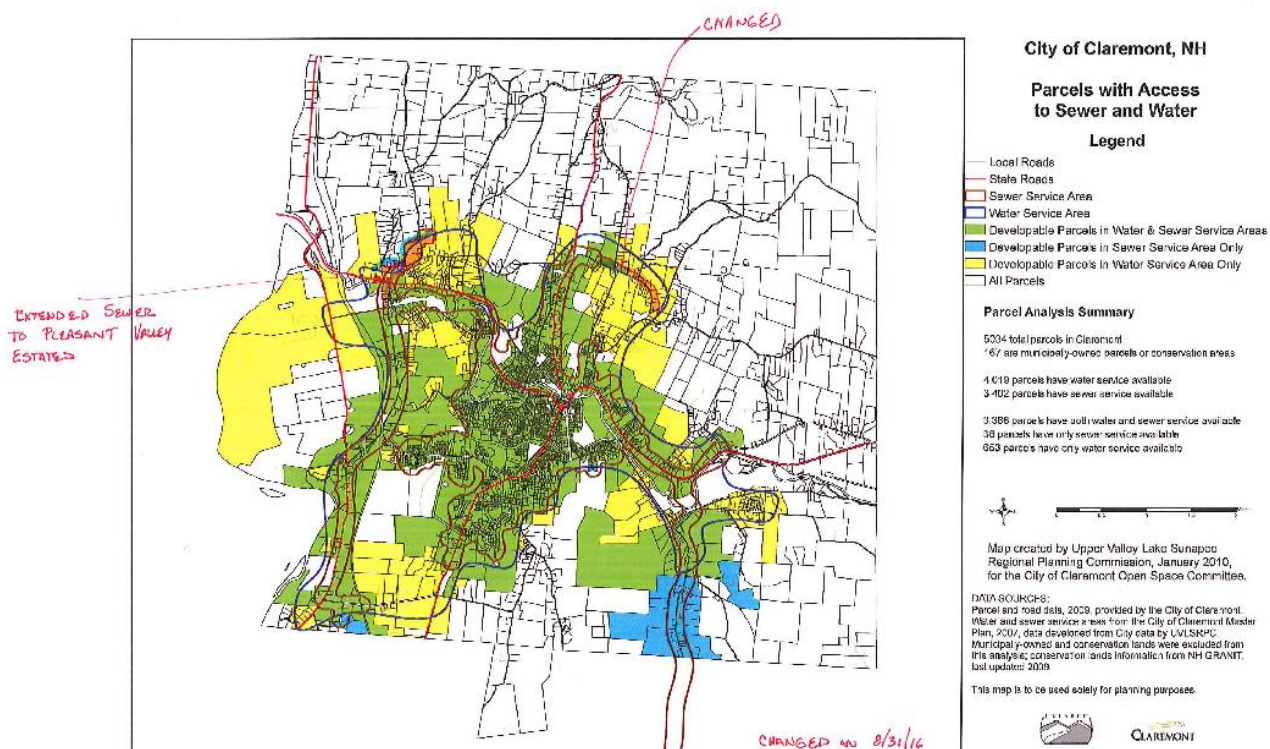


Figure 3.1 Map of the service area for the Claremont Water Department

The Claremont water system is supplied by three surface water sources: Whitewater Reservoir, Rice Reservoir and the Sugar River. The Rice Reservoir is located at the intersection of Route 120 and Winter Street and has a storage capacity of 40 million gallons. The Whitewater Reservoir is 17 acres, straddles the boundary of the City of Claremont and the Town of Cornish, and is reported to have a capacity of approximately 150 million gallons. This reservoir provides 70% of the City's water supply. The capacity of the Whitewater Reservoir is in question because it is unknown whether the 1972 breach of the Upper Whitewater dam reduced the Reservoir's holding capacity. Whitewater Reservoir is the City's most pristine and valuable source of public drinking water.

A pump and main along Winter Street were installed in 1981 to draw water from the Sugar River as a source for either the water treatment plant or the Dole Reservoir. This source has always been intended to supplement the Dole Reservoir as needed. The watershed for the Sugar River drains from a large area including Lake Sunapee to the east. The municipalities of Newport and Sunapee are upstream from the intake of this source. Potential Sources of Contamination include industrial and municipal waste, landfill leachate, polluted stormwater runoff carrying VOCs from accidental spills of oils and industrial chemicals, agricultural runoff, contamination from malfunctioning septic systems, and road salt to name a few. The Pump station at Sugar River has a design capacity of 8.0 MGD (Upper Valley Lake Sunapee Council, 1991).

Water from the Whitewater Reservoir, Rice Reservoir and the Sugar River sources feed the Dole Reservoir, a 50 million gallon storage reservoir prior to entering into the water treatment plant. Water is fed from the Whitewater and Rice Reservoirs through 10 inch distribution mains into Dole Reservoir (Upper Valley Lake Sunapee Council, 1991). The City of Claremont owns 50 acres around Rice Reservoir and 23 acres surrounding Dole Reservoir. City of Claremont owns 471 acres around the Whitewater Reservoir.

All of the water that enters the water treatment plant is treated in stages: coagulation, settling, filtration, chlorination to remove pathogens (disease-causing organisms), pH control, and corrosion control. Coagulation, settling, and filtration remove impurities from the water and the carbon filtration addresses taste and odor concerns. A complete description of the water treatment plant and the treatment process can be found in the 1978 "Report Relative to Proposed Water Treatment Plant, City of Claremont, New Hampshire".

3.2 An Abbreviated History

The City's public water system has undergone many changes of sources and storage systems over the years. What follows is an abbreviated version describing some of the reservoirs used through time.

In 1887 the NH General Court granted a charter to fourteen citizens and their associates to establish the Claremont Water-Works Company. The Claremont Water-Works Company was incorporated in 1887 By John L. Farwell, Edward J. Tenney, Ira Colby, George H. Stowell, H.W.

Parker, John T. Emerson, Herbert Bailey, Daniel W. Johnson, C.B. Rice, Edwin Vaughn, Oscar J. Brown, Albert Ball, and George H. Stowell, 2d, “for the purpose of bringing water into the villages in Claremont in said state by subterranean pipes.” This private company built the system in 1887-1888 using wrought-iron cement-lined pipes.

In 1898 the Claremont Water-Works Company was taken over by R.A. Quimby, O.D. Quimby, E.J. Rossiter, R.M. Sanders, and James L. Rice. (Easter, 1941). In 1899 the town of Claremont was authorized by “An act to authorize the Town of Claremont to procure by purchase, or under the power of eminent domain, or to put in a water supply.” (March 1, 1899; Chapter 180). The City of Claremont purchased the water system from the Claremont Water-Works Company in 1902 (Easter, 1941).

During the early years, the system was fed by two brooks by gravity, had two impoundments (Straw Reservoir and Phelps Reservoir) with storage capacity of 2,750,000 and 3,250,000 gallons respectively, 1 storage tank with a capacity of 22,000,000 gallons and 12 miles of distribution mains (Baker 1888). The system experienced a drought and resulting water shortage in 1889.

In 1900, to address future water shortages twelve wells were dug to a depth of about 14 feet on Kelsey’s Island in the Sugar River (New Hampshire State Board of Health, 1916). This source of water was considered an auxillary source to be utilized as a last resort during a few months per year. Water is pumped directly into water mains and the overflow is directed to the Dole Reservoir. Even in 1916 it was noted that “this water has never been satisfactory, due to odor and taste and accompanying iron, the character has in recent years shown steady deterioration.” (New Hampshire State Board of Health, 1916). While these words were written over one hundred years ago, they remain relevant today. Furthermore, the author recommended “it would be good policy, so far as feasible, to utilize Whitewater Brook as the regular source by constructing a dam, a storage for Whitewater Brook water of some 50,000,000 gallons would thereby be secured.” (New Hampshire State Board of Health, 1916).

In 1913, Claremont voted to raise \$72,000 to construct the Dole Reservoir, a storage of 37,000,000 gallons (Easter, 1941). In 1922 the City of Claremont asked E.E Davis to conduct another survey to provide recommendations for the water supply. He recommended that a reservoir site at the intersection of proposed 10” and 12” proposed water lines. In 1923 the Town voted to raise \$75,000 to build a diversion dam and gate chamber in Whitewater Brook and lay pipe lines to Dole Reservoir. This construction resulted in the creation of the Upper Reservoir. In 1933 the City faced another water shortage. In 1934, the City voted to raise \$50,000 to build the Rice Reservoir, named after the City’s superintendent who served in that position from 1898 until his death in 1933. (Easter, 1941).

In 1943 the Whitewater Brook Upper Dam was raised to a height of 62 feet to increase the storage capacity. The reservoir had a capacity of 420 acre feet (or approximately 136 million gallons) and a width of 450 feet. The City constructed the Whitewater Brook Lower Dam in 1965

which had a capacity of 170 million gallons. The original combined storage at the Whitewater Reservoir System was approximately 280 million gallons. The Upper dam failed in May, 1972 due to heavy rains, resulted in a maximum discharge on the order of 2,500 cubic seconds. Until this failure the City received nearly all of its drinking water from the Whitewater Watershed. The flood caused a portion adjacent to the concrete spillway to give way. The breach was responsible for considerable erosion and damage that occurred along the spillway channel as well as on the downstream face of the dam. The estimated top width of the breach was 21 feet, while the average depth of the breach was estimated to be 20 feet. (Singh, 2013).

In 1972, after the failure of the Whitewater Brook Upper Dam, the City of Claremont hired Whitman and Howard, Engineers and Architects to conduct a study on the City's water supply. In 1973, Whitman and Howard engineers issued their "Report Relative to Water Facilities, Claremont, New Hampshire" which declared repair of the dam was economically unfeasible. They recommended creation of a water treatment facility. Their study also included future water demand projections based upon the 1969 report by Anderson-Nichols & Co, entitled "Public Water Supply Study-Phase One Report." Whitman and Howard predicted that a treatment plant with the capacity to process 4.0 million gallons per day would be adequate for 2020 population projections. The study used a safe yield of 1.1 MGD from Whitewater Reservoir and 1.4 MGD from the Sugar River (Upper Valley Lake Sunapee Council, 1991). In the 2020 projection it was anticipated that Claremont's population would climb to 17,500 and water demand would be 2.5 MGD. Today in 2020, Claremont's population is approximately 12,967 and water demand is approximately 1.1 MGD.

Year	Projected Population (Whitman & Howard Projections)	Actual Population	Projected Water Consumption (MGD)	Actual Water Consumption (MGD)
1980	14,500	14,557	1.3	
1990	15,000	13,947	1.5	
2000	15,900	13,122	1.7	
2010	16,900	13,347	2.0	
2020	17,500	12,967* (2018)	2.5	1.1 MGD

Table 3.1 Projected and actual population and water consumption for the Claremont Water Department.

In 1981 the Claremont Water Treatment Plant was built where it is currently located at 196 Winter Street in Claremont, NH. A clear well was constructed in 2009 with a 600,000 gallon capacity at the water treatment plant.

Citing the 1973 "Report Relative to Water Facilities, Claremont, NH" by Whitman and Howard, Inc. and based upon "current patterns of growth and development" the Upper Valley Lake Sunapee Council (1991) noted that "it is expected that the existing community water system

will continue to adequately serve the City for the foreseeable future.” In the past, water loss due to leaks and drought have impacted the quantity of water available from Whitewater Watershed. In 2016 the Claremont Water Department embarked on a campaign to control lost water. For example, 3,500 water meters with drive by technology were installed. These meters track water use every 15 minutes and help to pinpoint leaks. The Claremont Public Works Department has addressed issues with water loss and it is hoped that the Whitewater Watershed will be adequate to supply the City’s needs, only calling upon the Sugar River source for emergency backup purposes.

3.3 Potential Sources of Contamination for Water Resources in the Whitewater Watershed

Potential sources of contamination are those activities which have the possibility to negatively impact drinking water resources. A combination of methods was used to identify the potential contamination sources (PSOCs) within the Whitewater Watershed. These include: NH DES Source Water Assessment Report (SWAP), NH DES and Town GIS data, NH DES One-Stop web-based data retrieval system, and local knowledge from the Whitewater Watershed Advisory Committee.

NH DES Source Water Assessment Reports (SWAP), completed in 2007 provide information about potential contamination sources for each individual public water source. The categories of potential contamination sources are ranked by NH DES as having “Low”, “Medium”, or “High” risks. According to the SWAP report for the Whitewater Reservoir there were eleven potential sources of contamination and they were all ranked as “Low” threats. In some cases, “Low” meant no PSOCs present for that category. The NH SWAP report indicated that there were no PSOCs present in the Reservoir’s hydrological area of concern. There were no threats from transportation corridors such as highways or railroads and no application areas for pesticides. The SWAP report indicated that there were no septic systems within 500 feet of surface water. Less than 10% of the HAC is urban, and there is no agricultural land. (NH DES SWAP Report, 2007).

NH DES One-Stop provides information from databases on remediation sites, initial spill response sites, aboveground and underground storage tanks, and Resource Conservation & Recovery Act (RCRA) sites or hazardous waste generators. No such sites were identified within the Whitewater Watershed. This finding was confirmed by the Source Water Committee based on local knowledge and Granite State Rural Water Association through windshield surveys throughout the protection area.

Inventory of Potential Sources of Contamination

The Committee found that there were six general categories of potential sources of contamination or “areas of concern”:

1. Insufficient existing water quality data and information about the reservoir

2. Bacteria and viruses from wildlife
3. Potential sources of contamination from recreational activities.
4. Potential sources of contamination from forestry activities
5. Ensuring there is adequate drinking water quality and supply for the future
6. Lack of public awareness about and appreciation for the Reservoir and the Claremont Public Drinking Water System

INSUFFICIENT WATER QUALITY DATA AND INFORMATION ABOUT THE RESERVOIR SYSTEM

Chemical and biological water quality information in the Watershed has not been monitored. Key parameters include phosphorus, nitrogen, sodium, chloride, dissolved oxygen and temperature, transparency and algal abundance to name a few. Phosphorus is the most important nutrient in determining water quality and lake health. If there is too much phosphorus in a water body, clarity decreases, total organic carbon increases, oxygen concentrations are depleted which in turn may result in a toxic algae bloom. Implementing a Volunteer Lake Assessment Program (VLAP) is an affordable way to measure water quality parameters. The NHDES VLAP program trains volunteers how to measure water quality variables and loans scientific equipment.

The Whitewater Watershed does not have a Volunteer Lakes Assessment Program in place. A VLAP program can help to gather information about important chemical and biological criteria such as mentioned above. Monitoring of Cyanobacteria has not occurred either. Cyanobacteria can be monitored by taking photos (Bloomwatch mobile app) or by collecting water samples (Cyanoscope, a sample kit). Observations should be made on the downwind shore of the Reservoir.

In addition, the Lower Reservoir's bathymetry (underwater topography) precise volume, and flushing rate are unknown. Little is known about mean and maximum depth or trophic class. VLAP volunteers could borrow the NHDES bathymetric measurer to determine volume or perhaps staff from the NH DES Watershed Bureau could determine reservoir volume. It is estimate that to hire a professional to determine the Reservoir's bathymetry might cost approximately \$5-10,000. This information is necessary to develop models on existing conditions and prepare models on the potential effects of various land use change scenarios.

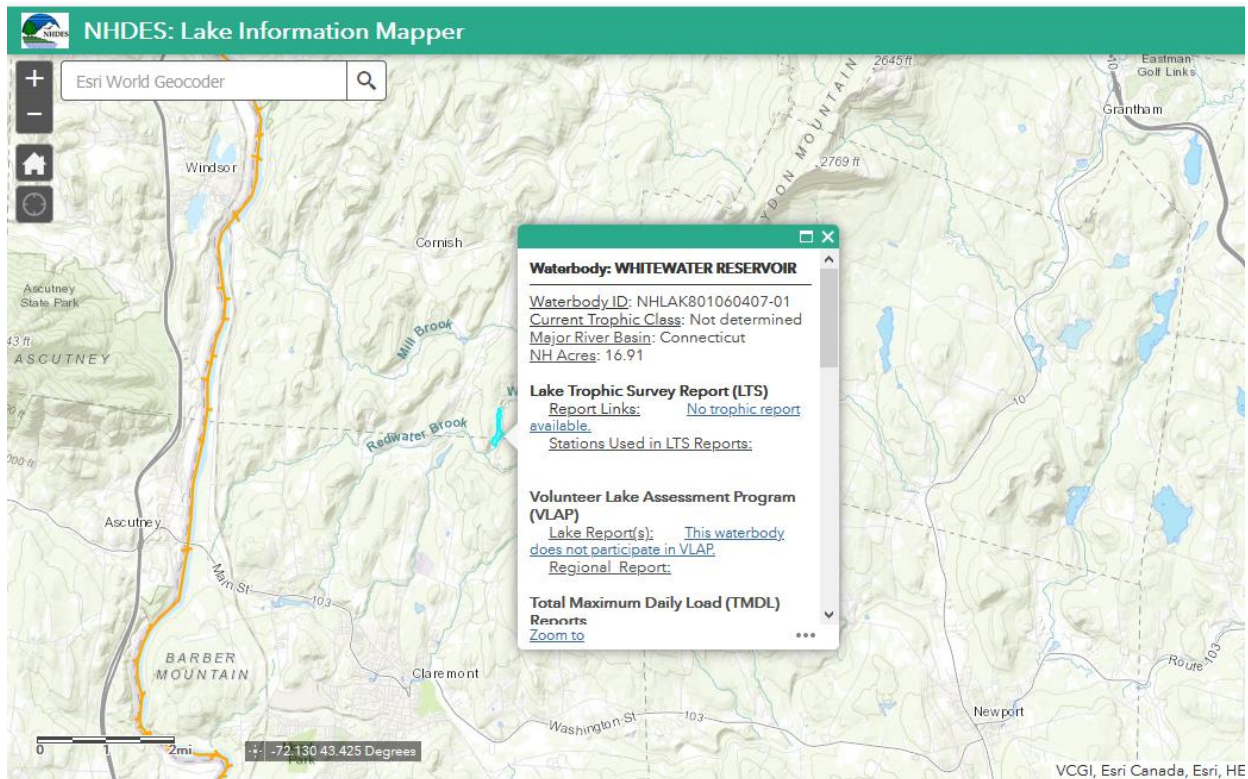


Figure 3.2 Map and information from the NHDES Lake Information Mapper demonstrating the need for information about the Whitewater Watershed.

POTENTIAL SOURCES OF BACTERIA AND VIRUSES FROM WILDLIFE

Fecal matter from geese and beaver may contaminate sourcewater with pathogens, bacteria or viruses. Some waterborne pathogens or diseases include typhoid fever, viral and bacterial gastroenteritis and hepatitis A. The presence of fecal contamination is an indicator that a potential health risk exists for individuals exposed to this water. The presence of beaver may pose a public health threat because they can be carriers of *Giardia Lamblia* and *Cryptosporidium* – pathogens identified within the Surface Water Treatment Rule. In addition, beaver may dam up and change the water level and flow potentially creating water quality and quantity issues.

There are significant geese populations around the Dole Reservoir. Geese population information for the Whitewater Reservoir is unknown. There has been intermittent beaver activity in the former Upper Reservoir area.

RECREATIONAL ACTIVITIES

There is a network of trails that is used by motorized recreation enthusiasts for snowmobiling, driving ATV's and 4x4s. The trail system is part of a larger trail network which extends throughout and beyond Sullivan County. Some of the trails are in good shape with only minor regular maintenance needed, such as cleaning of drainage ditches and creating additional water turn-outs. In other instances, bridges need repair and silt barriers are needed to divert water so

that it can percolate into the ground as opposed to going directly into a stream. The bridge on Skyline Drive is in disrepair because of the large volume of traffic from 4x4s. These trucks are too large for the current bridge. A larger bridge is needed to keep traffic out of the tributary and help prevent erosion. There are approximately 4-5 bridges within the Watershed that should be on a replacement schedule (Wilkie, 2020, Pers. Comm.) In some instances, trails need to be re-routed away from fragile ecological features such as in areas along the northern side of Whitewater Brook. The 2008 Forest Management Plan recommended that trails which “run through the buffer zone of the reservoir” should be “closed completely or relocated”. To date this has not occurred.

Trails need better signage describing trail locations, and use and respect for the land. Local club members are educated. ATV and Snow Rider members receive monthly email blasts which include educational information. However, only 50% of trail users are members of a NH club and registered with a local association. Many riders are from out of state and have less of a connection to protect local resources. Recreation users often shoot and destroy the signs. At times vehicle speed is an issue. There are fines for off trail riding and this activity is enforced by NH Fish & Game and Claremont police.

FORESTRY ACTIVITIES

Potential Sources of Contamination from forestry activities include risks of chemical spills from machinery, erosion concerns, and sedimentation in water bodies and tributaries. Forestry activities depending on scale can also result in increases of phosphorus that create undesirable changes in water chemistry. Too much phosphorus in a lake can increase total organic carbon, deplete oxygen and result in algal blooms. These factors can negatively impact water quality as a source of drinking water. As mentioned previously there are two streams which enter the Reservoir. Streams are an important source of water for the Reservoir, but they can also serve as conduits for pollutants, depending upon surrounding land use activities. It is important that any timber harvest activities do not negatively impact streams and other water resources.

ENSURING THERE IS ADEQUATE DRINKING WATER QUALITY AND QUANTITY FOR THE FUTURE

Currently the City of Claremont relies on the Whitewater Reservoir for 70% of its drinking water. The City also uses water from the Sugar River. The River source is vulnerable to spills, stormwater pollution and also provides water treatment challenges. Additional information is needed to assess the feasibility of re-developing the damaged Upper Whitewater Reservoir or create a storage system in the Redwater Watershed as additional sources.

A plan to acquire key parcels for water resource protection in the Watershed is lacking. Protection of key parcels, with the collaboration of willing owners through right of first refusal and a variety of conservation techniques such as fee simple purchase, conservation easements is needed.

LACK OF PUBLIC AWARENESS ABOUT AND APPRECIATION FOR THE RESERVOIR AND THE CLAREMONT PUBLIC DRINKING WATER SYSTEM

Watershed protection is the most successful when citizens understand the connection between land use and their drinking water supply. There are lots of opportunities to increase awareness about Claremont’s public drinking water system and the Whitewater Watershed.

Table 3.2 List of Challenges and Potential Sources of Contamination in the Whitewater Watershed

Description	Potential Source of Contamination or Challenge	Notes	Ranking	PSOC Distance from Reservoir
Insufficient water quality data and information about the reservoir	Lack of data	Data/information is needed to improve decision-making	High Priority	N/A
Nonpoint source pollution	Wildlife: geese and beaver	Pathogens, bacteria, viruses	High priority	Close
	Recreational Land Use	Volatile organic chemicals, sedimentation, turbidity	High Priority	Variable
	Forestry Activities	Volatile organic chemicals, sedimentation, turbidity	High priority	Variable
	Residential Land Use	Pathogens, bacteria, viruses, petroleum products from cars or oil tanks, household hazardous waste, lawncare products	Low	Far
Adequate future supply	Lack of information about additional sources	Can the Upper Reservoir be restored? Does it make sense to develop the Redwater Watershed as a source? Are there other sources	High	N/A
	A specific plan to acquire key parcels for water resources protection does not exist.		High	N/A
	Sugar River is a problematic source	The River source is vulnerable to spills and	High	N/A

		contaminated stormwater runoff. Water from this source poses treatment challenges.		
	Status of Claremont-owned properties	Deed research needed to determine if these properties in the Watershed have restrictions that protect the land in perpetuity	High	N/A
	Alarm needed for Reservoir dam	Alarm is needed in the event of emergency to notify residents downstream of dam. Water Department conducts dam inspections every 2 weeks, State inspects dam every 2 years.	High	N/A
Public Awareness	Low level of awareness	Low awareness often leads to lack of protection and complacency	High	N/A

The Whitewater Watershed Advisory Committee developed recommendations to address the potential sources of contamination list above. These recommendations are located in Chapter Five.

Chapter 4. Existing Water Resource Protection

4.1 Land Conservation

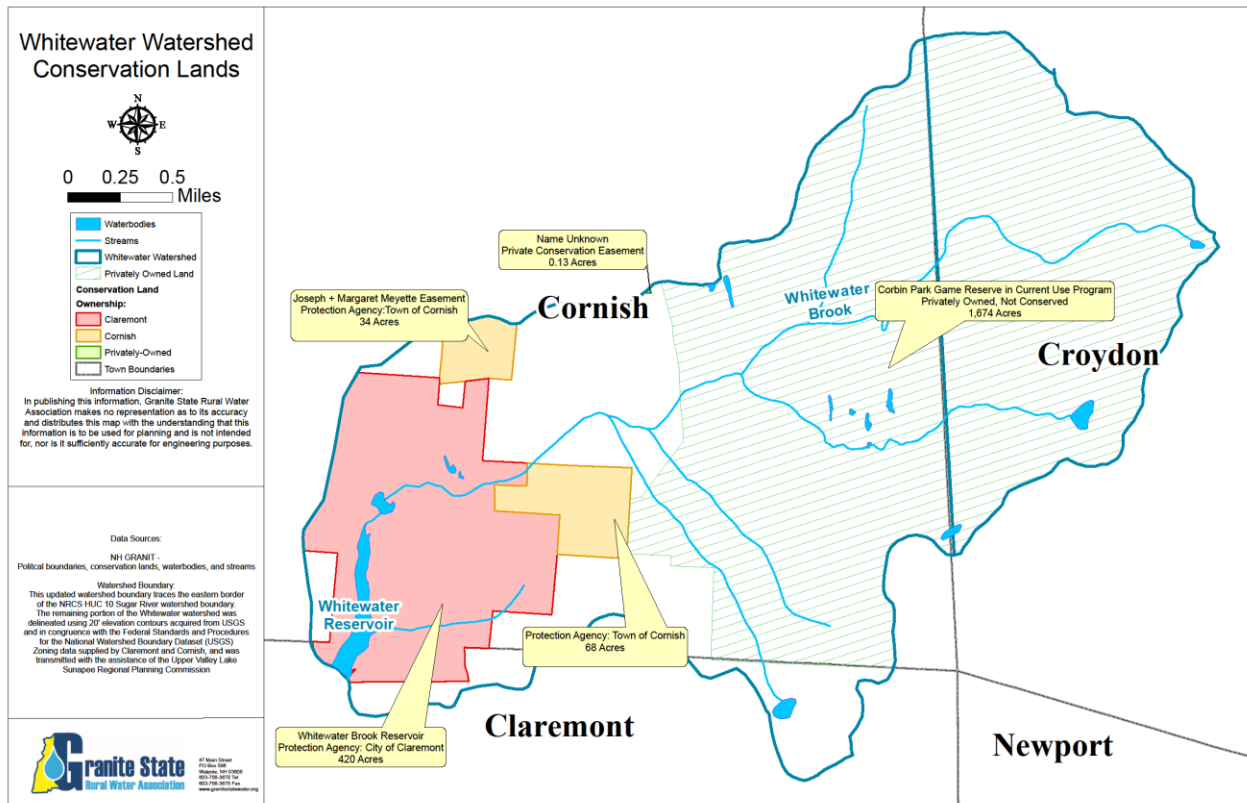
“Land conservation” occurs in a variety of forms. Land ownership for the purpose of conservation is one of the most effective ways to protect water resources. Land ownership through fee simple ownership or conservation easements provides the most control over land use activities. Fee simple ownership refers to complete ownership of all of the “bundle of rights” associated with a property. One way to ensure that land purchased for the purpose of water resources protection is protected in perpetuity, is to place a deed restriction or “restrictive covenant” on the property. For example, the deed restriction can require that the land remain undeveloped and be managed for the purpose of water quality protection in perpetuity. These legal restrictions “run with the land” from property owner to property owner. Another example is a conservation easement. A conservation easement is a permanent legal agreement between a landowner and a public agency or private nonprofit conservation organization. Conservation easements can limit or restrict how a property can be used. By placing a conservation easement on a property, the landowner transfers some of the development rights, to a responsible third party, such as a land trust whose mission is to safeguard the property for the purpose of conservation. The easement-holding entity is responsible for ensuring that the easement restrictions are met.

The City of Claremont owns five parcels of land within the Watershed. All of these parcels except for lot 12-38 in Cornish are located adjacent to the Reservoir. All of these parcels were purchased for the Claremont Water Department. It was outside the scope of this project to research whether these properties have restrictive covenants.

There are three parcels in the Watershed which have conservation restrictions. A least two of the easements are monitored by the Town of Cornish. Parcel 11-39 is a total of 179 acres of which 34 acres are located in the watershed. The second, lot 12-44 is a 68 acre parcel and is located entirely in the Watershed. Lastly there is a lot which intersects with the Watershed boundary and has .13 acres in the watershed.

Another example is the Blue Mountain Forest Association property. The Association has been a land owner in this area going back to the late 1800’s and has a history of cooperation with the City of Claremont. For example, during the drought of 1963, the Board of Directors of the Association voted to allow the City of Claremont to pump water from Governor and Sumner Ponds located on the Game Reserve. The entire reserve is enrolled in the Current Use Program reflects a long-term dedication to this property. The Current Use Program has been extremely successful in maintaining open space throughout New Hampshire. In order to qualify, a land owner must provide documentation of a certified tree farm, a Forest Stewardship Plan from a licensed forester or have a Forest Stewardship Plan developed privately. Yet despite being in

the Current Use program, a property owner can remove its land from the program with a tax penalty, and develop or sell any portion of its property. This means that any or all of the Association's 1,674 acres could be sold or developed in the future. According to the Association's current superintendent, Allen Welch, Blue Mountain Forest Association has no plans to sell or change its operations in the near future (Pers. Comm Allen Welch October 2019).



Map 5.1 Map of conserved properties in the Whitewater Watershed.

4.2 Municipal Master Plans

A municipal master plan is designed to guide the future actions of a community. It presents a vision for the future, with long-range goals and objectives for all activities that affect the local government. This includes guidance on how to make decisions on public and private land development proposals, the expenditure of public funds, availability of tax policy (tax incentives), cooperative efforts and issues of pressing concern, such as farmland preservation or the rehabilitation of older neighborhoods areas. Most plans are written to provide direction for future activities over a 10 to 20 year period after plan adoption. A master plan is the foundational document that provides the underlying support for a municipality's zoning ordinance.

The Cornish Master Plan stresses protection of the Whitewater Watershed. For example, the Cornish Master Plan states:

“The City of Claremont’s primary water supply is the Whitewater Brook Reservoir located in southeastern Cornish. Although protected by conserved lands, Cornish should continue to respect this water source in its land use decisions” (2009 Cornish Master Plan, Page 90). The Master Plan specifies on pages 90-91:

- Continue to protect the Whitewater Brook Reservoir
- Consider overlay districts to protect existing and any future public water supply.
- When considering land conservation, rank those lands which help to protect water quality as a priority.
- Avoid the development of any Potential Contamination Source near any existing or future public drinking water supply. Consider prohibiting these land uses within zoning. The NH Groundwater Protection Act identifies potential contamination sources (RSA 485-C:7. Prohibiting land uses such as salt storage, junk yards, underground storage tanks etc. is one of the most effective ways to protect water resources.

It is important to note that while the Master Plan refers to the Claremont properties as “conserved lands”, it is unclear whether restrictive covenants have been placed on these parcels.

The Cornish Master Plan recommends protection of wetlands within the Town, of which there are many wetland systems in the Watershed.

“Initiate a process for designation of Prime Wetlands as provided for in RSA 483A:7 that are based upon the *Guide to the Designation of Prime Wetlands in New Hampshire, 1983*. Such a designation will provide added protection for particularly valuable wetlands over and above the Wetlands Conservation District” (2009 Cornish Master Plan).

The Cornish Master Plan highlights that while Corbin Park, a private hunting and game preserve, has a considerable amount of land within Cornish, it is frequently taken for granted that this land is permanently protected and will never be developed. The land owned by the Blue Mountain Forest Association, is not permanently protected with either conservation easements or development restrictions. The master plan states that that areas adjacent to the park, such as the Whitewater Brook Reservoir should be considered for conservation as well (2009, Cornish Master Plan, Page 40). “Given the size of these lands, the impact from possible development of these areas is a concern. One option to help protect these lands is to consider

creation of a conservation zone” (2009, Cornish Master Plan Page 40). Map V-2 from the Town’s Master Plan depicts a conceptual future land use plan which includes a conservation corridor adjacent to the City of Claremont owned parcels in the Whitewater Watershed.

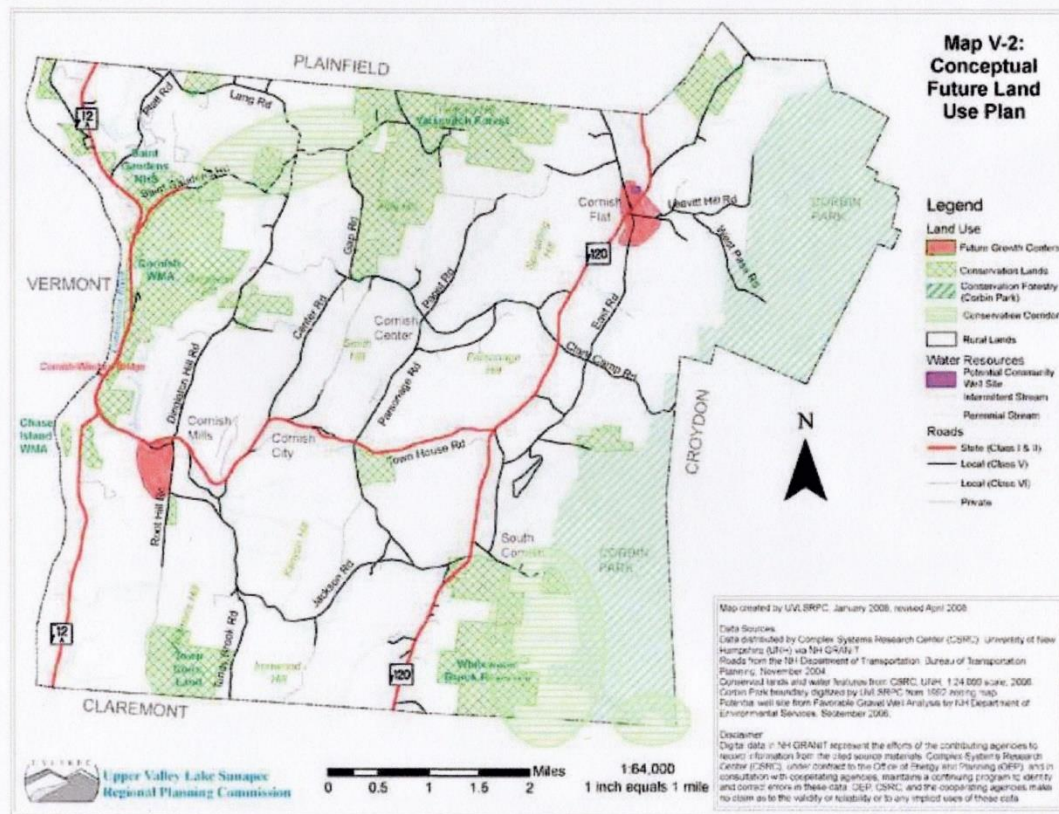


Figure 5.2 Map V-2: Conceptual Future Land Use Plan from the 2009 Cornish Master Plan.

In Chapter Seven of the Claremont Master Plan, in the “Water Resources Goals” Watershed protection is highlighted as a means of “protecting surface waters by managing the entire watershed that drains into it.” (Chapter 7: Natural Resources Page 7-8) and “is the preferred method” (Chapter 7: Natural Resources Page 9). The Claremont Master Plan identifies tools for Watershed Protection/Management including watershed-based zoning, overlay zoning, large lot zoning, and transfer of development rights. Other tools identified in the master plan include land acquisition from willing owners, aquatic buffers, better site design, erosion and sediment control and watershed stewardship programs.

The City of Claremont Master Plan (2017) specifically highlights the following objectives:
Objective No 1.1: Integrate water resource protection best practices into the City land use planning priorities and regulatory updates.

Action items:

1. Create and adopt a watershed management plan to regulate land use activities that affect the quality and quantity of the City's surface and groundwater resources.
2. Update the 1991 Source Water Protection Plan. Assess the current level of resource protection in zoning, subdivision and site plan regulations.

Objective No 1.2 Manage City-owned properties to protect water resources.

3. Enhance undeveloped buffer zones around City public water supply reservoirs.
4. Develop a land management plan for all City properties with an emphasis on water quality improvement and protection.

These two documents, the master plans for the communities of Claremont and Cornish serve as the foundation for the importance of the protection, conservation and management of water resources in the Whitewater Watershed.

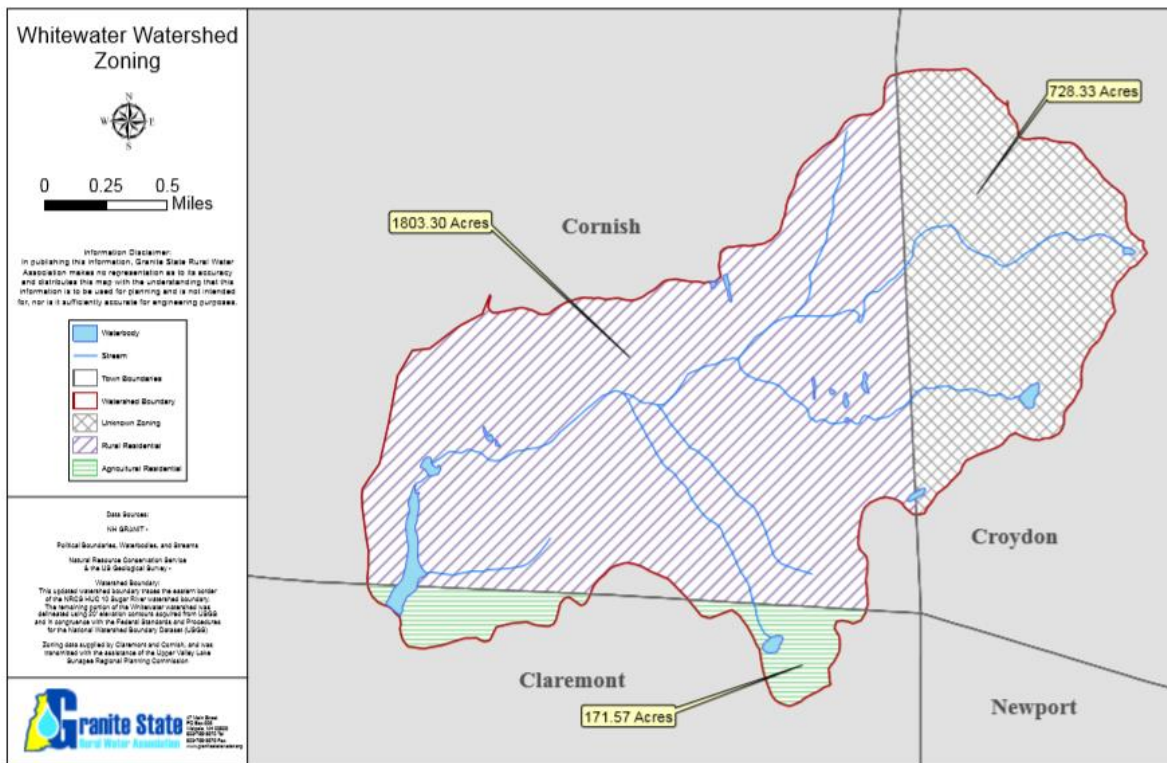
4.3 Regulatory Protection

Local regulations help protect water quality by directing development away from ecologically sensitive areas, by guiding the location of construction and development projects, and by prohibiting high risk land uses in specific areas. Local regulations include zoning bylaws and ordinances, subdivision regulations, local health ordinances, and site plan review regulations.

Zoning within the Whitewater Watershed

Zoning regulates land use including the size, shape, and permitted uses of lots and structures. Zoning controls where people live and work. The purpose of this land use control mechanism is to separate incompatible land uses in order to protect the public from health risks and to guide development to appropriate districts.

Ideally zoning regulates land use in order to meet the goals set forth in a municipality's master plan. In a master plan, a community describes its vision for the future. It is the zoning regulations that help to implement this vision. For the purpose of this Action Plan, zoning ordinances in the Watershed towns were reviewed for characteristics of water resource protection. This review was not a qualitative review for effectiveness but rather a check for presence or absence. The entire Watershed is zoned as Rural Residential or Agricultural Residential, promoting larger lot sizes for residential use.



Map 5.3 Zoning within the Whitewater Watershed is rural residential and agricultural residential.

In addition to Rural Residential, the Town of Cornish has two overlay districts, both of which are designed to protect water resources. The Town of Cornish has a Wetland Overlay District. The District is defined as all areas comprising bogs, marshes, swamps and other poorly drained or poorly drained soils together with a 100-foot buffer zone around such areas. The objective of this district is to limit development in and immediately around wetlands to protect those wetlands because they provide floodwater storage, wildlife habitat and groundwater recharge. They also serve to protect and enhance surface water quality through filtration and augmentation of stream flow during dry periods (Town of Cornish Zoning Ordinance, 2015, Page 11). The Overlay Districts are defined in relation to the edge of the wetland or shoreline. Since shorelines and wetlands are dynamic in that their location can change over time, the location of the overlay district also can change. For this reason, the Wetland Overlay Districts are not shown on the Cornish Zoning Map. According to a Cornish Town official, the definitive way to make a finding regarding the location of an overlay district is to have an area surveyed by a wetlands or other appropriately certified engineer. This determination occurs during subdivision proceedings and occasionally building permits. In Planning Board actions, the onus is on the landowner to show the location of the overlay district.

The Town of Cornish also has a Shoreland Conservation Overlay District. “The Shoreland Conservation Overlay District is defined as comprising all lands within 100 feet of the shore of all water courses and water bodies. The 100-foot distance shall be measured horizontally from the top of the bank above the water course or water body in question. The Town’s water courses and water bodies are vital natural resources providing significant scenic, recreational, and wildlife values with the potential for public water supply. The primary objective of this District is to preserve and protect the Town’s water courses and water bodies by permitting only those uses which can be harmoniously, appropriately and sagely located adjacent to water course and water bodies from sedimentation, turbidity, runoff of stormwater and effluent from sewage disposal systems, preserve scenic views, wildlife habitat and vegetative cover.” (2015 Town of Cornish Zoning Ordinance, Page 12).

[RSA 483-B:4 The Shoreland Water Quality Protection Act](#)

The Shoreland Protection Act provides minimum protective standards for activities occurring within 250 feet of lakes and ponds with a surface area of 10 acres or more. The Whitewater Reservoir is listed on the consolidated list of water bodies that are subject to the Shoreland Water Quality Protection Act.

https://www.des.nh.gov/organization/divisions/water/wetlands/cspa/documents/consolidated_list.pdf

[Administrative Rule ENV Dw 902.11](#)

Nearly all water bodies that serve as public drinking water supplies have watersheds which extend beyond municipal boundaries. This makes protecting these resources challenging. In 1898, a law was passed which gave the NH Board of Health the authority to develop administrative rules to protect water supply watersheds. RSA:23 gives NH DES the authority to adopt rules to protect active surface water sources. ENV-Dw 902.11 Protection of the Purity of Whitewater Brook, Rice Reservoir and Dole Reservoir and their Watersheds was last revised in 2014 (See Appendix B). Some of the restrictions specified in this rule include:

- (1) A person shall not build, continue or maintain a structure of any kind in which animals or fowl are kept 75 feet of said book or reservoirs or any inlet thereof or tributary thereto;
- (2) A person shall not allow wastes or waters that have been used for washing or cleansing materials, persons, or food to run into said brook etc.
- (3) A person shall not throw or deposit any dead animal or fish or parts thereof, decayable material, manure, or human wastes into said brook or reservoirs
- (4) A person shall not throw any sawdust or allow any sawdust to fall into said brook or reservoirs
- (5) A person shall not trespass, boat, bathe, swim, fish, or carry on any activity whatever whether recreational, occupational or other nature in the waters or on the ice of Whitewater Brook Reservoir and Intake Reservoir located on Whitewater Brook

(6) A person shall not throw, deposit, or allow to remain on the ice of the waters of said brook or reservoirs any matter, waste or materials such as described in (2), (3), and (4), above.

(h) The city of Claremont shall post a summary of the prohibitions contained (g) above, at all public access locations where persons might reasonably be expected to access Whitewater Brook, Rice Reservoir, or Dole Reservoir or their watersheds or tributaries.

It is important to note that these protective regulatory measures need to be implemented and supported by local municipalities in order to be effective.

Federal and State Surface Water Quality Standards

The establishment of water quality standards is one of the key components of the federal Clean Water Act, setting the desired water quality goals to be met by the state. The State of New Hampshire has numerous statutes and rules that are designed to protect lakes. The laws are based on the philosophy that it is "easier, cheaper, and more logical to protect lakes from degradation than it is to restore degraded lakes" (State of New Hampshire 2000 Section 305(b) Water Quality Report). Water bodies created by impoundments are also referred to as "lakes" (NH Water Resources Primer, 2008).

Water quality standards can be defined as specific provisions of state or federal law that are adopted to "protect the public health and welfare, enhance the quality of the water, and serve the purposes of the Clean Water Act." Water quality standards ensure the physical, chemical, and biological integrity of the New Hampshire's waters are maintained and provide for the protection and propagation of fish, shellfish, wildlife, and recreation that takes place in and on the water. Federal water quality standards require states to designate various uses to their waterbodies, which in turn determine the level of water quality to be achieved in order to meet the goals of the Clean Water Act. New Hampshire defines these designated uses by classifying waterbodies.

Since 1991, the surface waters of New Hampshire have been classified by the state legislature (RSA 485-A:8) as either Class A or Class B. Class A waters are considered to be of the highest quality and considered optimal for use as water supplies after adequate treatment. Sewage discharges are prohibited in these waterbodies. Class B waters are considered acceptable for fishing, swimming, and other recreational purposes, and for use as water supplies after adequate treatment has been applied.

Waterbody classifications are supported by establishing numeric and narrative criteria. Numeric criteria are specific measures of water quality that are considered scientifically sound in order to protect the designated use of the waterbody/segment. These usually include parameters such as dissolved oxygen, temperature, pH, metals, and toxic pollutants. Narrative criteria apply to all designated uses and are statements that describe the desired level of water quality to be achieved, such as "all waters shall be free from substances in kind or quantity which settle to

form harmful deposits, float as foam, scum, debris, or other visible substances, interfere with recreational activities..."

The final component of New Hampshire's Water Quality Standards are specific provisions established to ensure that degradation of existing beneficial uses and the level of water quality necessary to protect the existing uses are maintained and protected. These antidegradation provisions apply to such things as new or increasing point and nonpoint discharges of pollutants, alterations to the hydrology of a system caused by dams or flow diversions, and all activities that would lower water quality and affect the beneficial uses. Provisions are established for Class A, and Class B Waters. (NHDES <https://www.des.nh.gov/organization/divisions/water/wmb/wqs/history.htm>)

For a full listing of pertinent statutes established to protect New Hampshire lakes see Appendix C.

Water Quality Monitoring Programs

NHDES has several ongoing surface water monitoring programs. These programs determine if water quality standards are being met through routine monitoring activities, provide educational opportunities to the general public, and conduct research and abatement practices on known pollutant sources.

Ambient Monitoring Program: The ambient sampling program monitors approximately one hundred sites annually, collecting samples once a month during seasonally low streamflows in June, July, and August. Sites are rotated among the 5 major river basins. Additionally, 12 Primary Monitoring Network trend sites, and five National Water Quality Surveillance System sites are sampled every year and have been since 1974. Dissolved oxygen, temperature, conductivity, pH, turbidity, ammonia, phosphorus, total kjeldahl nitrogen and nitrate nitrogen, and E. coli bacteria are measured during each visit. Other parameters sampled at least once a year are metals (Al, Cu, Pb, Zn), hardness and alkalinity. Various parameters have been sampled historically and have either been discontinued or are monitored less frequently.

Volunteer monitoring: Volunteer monitoring is an integral component of NHDES monitoring efforts, providing data coverage of lakes and rivers to enhance water quality information in New Hampshire. The Watershed Management Bureau coordinates the Volunteer Lake Assessment Program (VLAP). This program provides education, technical assistance, and equipment loans to enable citizens to be involved in protecting and monitoring their watershed.

Biomonitoring Program: This program was established to help fulfill the requirements of the Clean Water Act, "maintaining the biological integrity of the nation's waters." Aside from monitoring the resident aquatic communities in state waters for support of the aquatic life use designation, routine chemical parameters are monitored to complement the biological data. Dissolved oxygen, temperature, pH, conductivity, acid neutralizing capacity, and total dissolved

solids are measured at each monitoring site and are usually taken several times at the same location through the course of the summer. Other parameters are measured depending on the specific nature of the site, site history, and determined need.

Chapter 5. Recommendations for Protecting Water Resources in the Whitewater Watershed

Watershed protection is best achieved through a portfolio of approaches. Choices about which approaches to implement are best made at the local level. Local advisory committees have a better understanding of the issues and the solutions. There are at least eight categories of tools to achieve watershed protection. These include:

1. Land Use Planning
2. Land Conservation
3. Aquatic Buffers
4. Better Site Design
5. Erosion and Sediment Control
6. Stormwater Best Practices
7. Non-Stormwater Discharges
8. Watershed Stewardship Programs

This chapter summarizes the recommendations identified by the Whitewater Watershed Advisory Committee. Specific recommendations are listed as objectives and have more specific concrete tasks or “strategies” so that they may be achieved.

Each of the Whitewater Watershed Advisory Committee’s recommendations have been classified under these four general goals.

Goal #1: Promote Watershed management decision-making based upon sound scientific information.

Goal #2: Reduce potential pollution from nonpoint sources.

Goal #3: Ensure that the City of Claremont continues to have an adequate supply of clean, safe drinking water in the future.

Goal #4: Increase appreciation for the Whitewater Watershed and the Claremont Public Drinking Water Supply

Each goal is further divided into more specific objectives and even more specific strategies. Strategies are specific actions which can be implemented in order to meet the objective. The overall emphasis is on maintaining or improving water quality by identifying specific issues and tasks.

The objectives and strategies are listed in a “report card” format which specifies potential lead agencies, partners, funding sources, timeline and benchmarks. When it comes time to review progress of this Action Plan, this format can aid in the evaluation process. Goals, objectives and strategies are presented in a narrative in this chapter and also summarized in Table 5.1.

Goal: (to maintain or improve water quality)

Objective: (To implement effective practices that will meet this goal).

Strategy: (The specific practice that can be implemented to achieve the objective).

Potential: Lead Agency and Partners: describes a likely candidate for implementing the strategy and highlights potential partners.

Potential Funding Sources: Identifies potential sources for financial support

Time Frame: The anticipated time it will take for either implementation of the strategy or to bring about the desired outcome.

Benchmark: The desired outcome of the strategy.

Figure 5.1. Report card format of goals, objectives and strategies.

Goal # 1: Promote Watershed management decision-making based upon sound scientific information

During the course of this planning process the Committee found a lack of sufficient scientific data about the watershed including water quality in the reservoir and its tributaries, statistics about the Reservoir ranging from the lack of a bathymetric map, the water body’s tropic level, and flushing rate to name a few. Data is absent for the Reservoir’s chemical and biological parameters. For example, data about pH, phosphorus concentrations, alkalinity, dissolved oxygen is unavailable. It is not possible to make sound decisions without this data. The Committee developed one objective and 4 strategies to meet this goal.

Objective #1:

Increase scientific knowledge and understanding of the Whitewater Watershed through research and monitoring.

Strategies

Implement a Volunteer Lakes Assessment Program in the Watershed that collects water quality chemical and biological parameters. The NH DES VLAP program trains volunteers how to measure water quality variables and loans equipment. At the very least P samples should be taken after ice out because the reservoir will be mixed at that time. Also take P samples in June, July, August, September and once around the second week of October. Samples need to be

sent to a lab that can measure for very small concentrations. The Colby-Sawyer lab is equipped to process these samples and a relationship could be created with the college.

- **Potential Lead Agencies:** Claremont Water Department, Conservation Commissions from Claremont, Cornish and Croydon, Colby-Sawyer Lab, NH DES Volunteer Lake Assessment Program, NH DES Watershed Bureau
- **Potential Funding Sources:** Claremont Water Department, NH DES Sourcewater Grant, NH DES VLAP Program
- **Timeframe:** Begin July 2020
- **Benchmark:** A VLAP program has been created and volunteers are collecting data.

A. Conduct studies to determine the reservoir's morphometry, create a bathymetric map, determine reservoir volume, trophic level and flushing rate.

- **Potential Lead Agencies:** Claremont Water Department, NHDES, VLAP Program
- **Potential Funding Sources:**
- **Timeframe:** Begin July 2020 to develop an RFP for services
- **Benchmark:** The above information will be available.

B. Inspect the Whitewater Reservoir for presence of cyanobacteria blooms using aids such as the Bloomwatch app or Cyanoscope sampling kit.

- **Potential Lead Agencies:** Claremont Water Department, NH DES Watershed Bureau, NH DES Groundwater and Drinking Water Bureau
- **Potential Funding Sources:**
- **Timeframe:** Begin June 2020
- **Benchmark:** Periodic monitoring of cyanobacteria is occurring.

C. Conduct a build-out analysis of the Watershed to determine the highest rate of development possible.

- **Potential Lead Agencies:** Watershed Municipalities, Upper Valley Lake Sunapee Regional Planning Commission
- **Potential Funding Sources:** NH DES Regional Environmental Planning Program
- **Timeframe:** 2021
- **Benchmark:** A build out analysis will have been completed for the Watershed.

Goal # 2: Reduce Pollution from Nonpoint Sources

Objective #1: Reduce nonpoint sources of bacteria and viruses.

Strategies

- A. Install coyote decoys to deter geese population at the Dole Reservoir. Geese population fluctuates seasonally from about 40 to 300 geese. Geese are a potential source of bacteria and viruses due to fecal coliform. Determine if decoys are also warranted at the Whitewater Reservoir. Geese populations may also be controlled by harvesting by the Claremont Water Department.**
- **Potential Lead Agencies:** Claremont Water Department
 - **Potential Funding Sources:** City of Claremont
 - **Timeframe:** In progress
 - **Benchmark:** Geese droppings will be lessened surrounding reservoirs and geese populations will be lowered.
- B. Monitor beaver activity and address potential issues with appropriate measures.**
- **Potential Lead Agencies:** Claremont Water Department, Claremont and Cornish Conservation Commissions, Beaver Expert Skip Lisle from Vermont).
 - **Potential Funding Sources:** City of Claremont
 - **Timeframe:** Ongoing
 - **Benchmark:** Water quality issues created by beaver are kept to a minimum.

Objective #2: Reduce nonpoint source potential pollutants (erosion, sediments, turbidity and VOCs) from recreational trails.

Strategies

- A. Develop a rotational repair schedule for bridges in the Watershed. Immediately repair the bridge approach on Skyline Drive (Class VI road) and direct water away from the tributary.**
- **Potential Lead Agencies:** Sullivan County ATV Club/Sugah Valley Snow Riders, with letter of support from City of Claremont
 - **Potential Funding Sources:** Department of Natural and Cultural Resources – Bureau of Trails
 - **Timeframe:** April 2020, In progress
 - **Benchmark:** Bridge will be repaired to support recreational traffic and stormwater flow will be redirected away from tributary.
- B. Implement limited speed zones and re-route trails that are close to water resources and other sensitive ecological features. Develop maps and signs which show where these limited speed zones are located and share them with recreational users.**
- **Potential Lead Agencies:** Sullivan County ATV Club/Sugah Valley Snow Riders, with Claremont Water Department.

- **Potential Funding Sources:** NHDES Sourcewater grant
- **Timeframe:** Submit NHDES grant application November 2020; Install signage in 2021
- **Benchmark:** Educational materials will be created and shared. Signs will be posted.

Objective #3: Reduce nonpoint source potential pollutants (erosion, sediments, turbidity and VOCs) from forestry activities on Claremont-owned properties in the Watershed.

Strategies

- A. Develop a new forest management plan for the Claremont City-owned Watershed properties which highlight preservation of water quality has its most important goal.
 - **Potential Lead Agencies:** Claremont Directors of Parks & Recreation, Claremont Conservation and Cornish Conservation Commissions.
 - **Potential Funding Sources:** City of Claremont
 - **Timeframe:** July 2020 put out an RFP
 - **Benchmark:** A new forest management plan is created which specifies that protection of the Whitewater Reservoir is City's primary goal.

- B. On the Claremont-owned parcels, no harvest zone buffers should be established around the Reservoir and tributaries. These areas will be surveyed and demarcated on the ground and will have accompanying detailed GIS maps. The buffers will be compatible and supported by the Shoreland Water Quality Protection Act, and the Cornish Wetland Overlay Districts and the Cornish Shoreland Overlay District regulations. According to Limnologist Don Kretchmer, at the very least "There should be 300 foot buffers around surface water bodies and 100 foot buffers next to tributaries in the event that timber harvest takes place on City-owned parcels".
 - **Potential Lead Agencies:** City of Claremont, Town of Cornish, Conservation Commissions of the Watershed Municipalities.
 - **Potential Funding Sources:** City of Claremont
 - **Timeframe:** 2021
 - **Benchmark:** Buffer zones are demarcated prior to any harvest on Claremont-owned parcels.

- C. Evaluate the potential effects of various scenarios of timber extraction on City-owned properties in the watershed before any more harvesting takes place. Hire professionals with forestry expertise and knowledge in limnology to assess the potential effects of that disturbance from forestry activities might have on Phosphorus concentrations and other key parameters.

- **Potential Lead Agencies:** City of Claremont, Claremont Water Department, Claremont Directors of Parks & Recreation, Claremont Conservation and Cornish Conservation Commissions.
- **Potential Funding Sources:** NH DES Source Water Grant program
- **Timeframe:** July 2020 begin to develop grant application, submit grant application November 1, 2020
- **Benchmark:** Report with accompanying model will be completed and provide assessment of forest management scenarios.

D. Notify the Claremont Water Department when a timber harvest is scheduled to occur in the Watershed.

- **Potential Lead Agencies:** Watershed municipalities, Claremont Water Department, Claremont Directors of Parks & Recreation, Claremont, Conservation and Cornish Conservation Commissions.
- **Potential Funding Sources:**
- **Timeframe:** Ongoing
- **Benchmark:** Claremont Water Department will be notified.

Goal #3: Ensure that the City of Claremont continues to have an adequate supply of clean, safe drinking water into the future.

Objective #1: Evaluate the adequacy of the quantity and quality of existing drinking water sources to meet future public drinking water needs.

Strategies

A. Assess options for additional sources of public drinking water. For example evaluate the feasibility of repairing the Whitewater Upper Reservoir or creating a storage system in the Redwater Watershed.

- **Potential Lead Agencies:** Claremont City Council, Claremont Water Department
- **Potential Funding Sources:**
- **Timeframe:** 2021
- **Benchmark:** Research will be completed with an accompanying report that provides recommendations for additional sources of drinking water.

B. Due to issues of water quality and treatment, phase out use of the Sugar River as a source of public drinking. The Sugar River source should be relegated as a source of last resort.

- **Potential Lead Agencies:** Claremont City Council, Claremont Water Department
- **Potential Funding Sources:**
- **Timeframe:** 2022
- **Benchmark:** The Sugar River will no longer be used as a source for Claremont's public drinking water system.

C. Continue the Claremont Water Department water conservation program (leak detection, water pipe cleaning) to limit water loss.

- **Potential Lead Agencies:** Claremont Water Department
- **Potential Funding Sources:** NH DES Leak Detection Grant
- **Timeframe:** Ongoing
- **Benchmark:** The system's water loss will be less than 15%.

D. Implement Claremont Water Department's Energy Conservation Project.

- **Potential Lead Agencies:** Claremont Water Department
- **Potential Funding Sources:**
- **Timeframe:** 2021
- **Benchmark:** The Energy Conservation Project will be completed.

Objective #2: Conserve and protect significant and sensitive water resources in the Whitewater Watershed.

A. Develop a plan to conserve key properties which protect water resources. Work with willing property owners and use a variety of conservation techniques ranging from conservation easements and land acquisition. Seek right of first refusal on key parcels.

- **Potential Lead Agencies:** Upper Valley Land Trust, Upper Valley Lake Sunapee Regional Planning Commission, Conservation Commissions, Claremont Water Department, Whitewater Watershed Advisory Committee
- **Potential Funding Sources:** NH Drinking Water Trust Fund, NH Land and Community Heritage Investment Program (LCHIP)
- **Timeframe:** Ongoing
- **Benchmark:** Significant ecological water resources features will be protected through fee simple acquisition, conservation easements, tax incentives and other funding techniques.

B. Watershed municipalities should adopt a watershed overlay district which follows the boundaries of the Whitewater Watershed and directs development away from ecological sensitive areas, guides construction and development, and prohibits high risk land uses.

- **Potential Lead Agencies:** Watershed municipalities, Whitewater Watershed Advisory Committee, Watershed Planning Boards Granite State Rural Water Association

- **Potential Funding Sources:** NH DES Sourcewater Water grant program
- **Timeframe:** 2020-2025
- **Benchmark:** A watershed overlay district will be adopted in the watershed municipalities.

C. Research whether parcels owned by the City of Claremont have a covenant or deed restriction that requires the land to be permanently conserved for drinking water purposes. If there is no restriction, put covenants on these forementioned parcels.

- **Potential Lead Agencies:** City of Claremont
- **Potential Funding Sources:** NH DES SW grant
- **Timeframe:** 2021
- **Benchmark:** Claremont owned parcels in the Watershed will have covenants that run with the land, requiring that these parcels be protected in perpetuity for the purpose of drinking water supply.

D. Continue the collaborative work of the Whitewater Watershed Advisory Committee. Watershed protection is most successful where local communities have active citizens who promote watershed protection and safeguard water quality. Water quality protection and improvements will occur through implementation of this plan and increasing awareness of watershed issues. The Advisory Committee is the best entity to oversee and ensure continued implementation.

- **Potential Lead Agencies:** Watershed municipalities, Whitewater Watershed Advisory Committee.
- **Potential Funding Sources:** Inkind, Watershed municipalities
- **Timeframe:** Ongoing
- **Benchmark:** Plan is implemented and updated into the future.

Goal #4: Increase appreciation and awareness about Claremont’s public drinking water system and the Whitewater Watershed.

Objective #1: Raise awareness about the Reservoir as the primary source of drinking water for the City of Claremont.

Strategies

A. Present the findings of this Action Plan to leadership, boards, and commissions of the watershed municipalities.

- **Potential Lead Agencies:** Whitewater Watershed Advisory Committee, Claremont Water Department, Granite State Rural Water Association

- **Potential Funding Sources:**
- **Timeframe:** June 2020
- **Benchmark:** Presentations will be made.

B. Distribute both electronic and paper copies of this Action Plan in all Watershed communities. For examples copies should be made available for boards and commissions, libraries, town halls. Electronic copies should be available on municipal websites and the NHDES source water protection website.

- **Potential Lead Agencies:** Watershed municipalities, Claremont Water Department, Granite State Rural Water Association.
- **Potential Funding Sources:**
- **Timeframe:** 2020
- **Benchmark:** Paper and digital copies will be widely distributed.

C. Install a kiosk at the northern entry to the Watershed that educates recreational users about the importance of the Watershed as the source of Claremont’s drinking water supply.

- **Potential Lead Agencies:** Claremont Water Department, Claremont and Cornish Conservation Commissions, Sullivan County ATV/Sugah Valley Snow Riders
- **Potential Funding Sources:** NH DES Sourcewater Protection grant
- **Timeframe:** Write grant application summer 2020, apply November 2020 for grant
- **Benchmark:** Kiosk is installed.

D. Install signage “Entering Drinking Water Watershed” at all points of entry for recreational trails.

- **Potential Lead Agencies:** Claremont Water Department
- **Potential Funding Sources:** NH DES Sourcewater grant program
- **Timeframe:** Write grant application summer 2020, apply November 2020 for grant
- **Benchmark:** Signs at points of entry are installed.

E. Engage schools in water resource education programs such as “ProjectWET” and tours of the Claremont Drinking Water and Wastewater Treatment Plants and the Whitewater Watershed.

- **Potential Lead Agencies:** Claremont Water Department, NH DES Groundwater and Drinking Water Bureau, Claremont schools
- **Potential Funding Sources:**
- **Timeframe:** Ongoing

- **Benchmark:** Claremont teachers use the ProjectWET curriculum and school children take tours of the water and wastewater treatment plant and Whitewater Watershed.

F. The Claremont Water Department will participate in the annual NH Drinking Water Taste Test contest, a fun contest which raises awareness about the importance of drinking water. This contest is held every September at Granite State Rural Water Association’s annual water operator conference.

- **Potential Lead Agencies:** Claremont Water Department, NH DES Groundwater and Drinking Water Bureau, US Environmental Protection Agency, USDA Rural Development, Granite State Rural Water Association
- **Potential Funding Sources:** sampling bottles provided in kind
- **Timeframe:** Ongoing
- **Benchmark:** Claremont Water System participates in the contest and uses it to raise publicity about their source of public drinking water.

G. Send letters to all property owners within the Watershed to inform them that their property is within a drinking water watershed and the ways that they can prevent potential sources of contamination. See letter in Appendix A.

- **Potential Lead Agencies:** Claremont Water Department
- **Potential Funding Sources:**
- **Timeframe:** Ongoing
- **Benchmark:** Using the list of parcels and property owners, letters will be sent on an annual basis. Prior to mailing, the list will be reviewed to make sure it is current.

H. Share the link for the video of the Whitewater Watershed Kickoff meeting filmed by CCTV. Share this video on “What’s up Claremont” and on the Claremont Conservation Commission FaceBook page as well as on “Connect Cornish”.

- **Potential Lead Agencies:** Claremont Water Department
- **Potential Funding Sources:**
- **Timeframe:** June 2020
- **Benchmark:** Links to the video will be shared to the platforms listed above.

I. Engage stakeholders, conservation commission members, planning board representatives and others from the Town of Croydon to be part of the Whitewater Watershed protection initiative.

- **Potential Lead Agencies:** Whitewater Watershed Advisory Committee, Claremont Water Department
- **Potential Funding Sources:**

- **Timeframe:** Ongoing
- **Benchmark:** Representatives will become part of the Whitewater Watershed Advisory Committee.

Table 5.1 List of Management Activities

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
Goal #1: Promote Watershed management decision-making based upon sound scientific information.	Objective #1: Increase scientific knowledge and understanding of the Whitewater Watershed through research and monitoring.	A. Implement a water quality monitoring program in the Reservoir and tributaries which examines water quality parameters such as P, dissolved oxygen, pH, Alkalinity, etc. Collect data by creating a Volunteer Lakes Assessment Program in the Watershed.	Claremont Water Department and Claremont/Cornish/Croydon Conservation Commission Colby-Sawyer College Lab NH DES Volunteer Lake Assessment Program NH DES Watershed Bureau	Claremont Water Department NH DES Sourcewater Grant NH DES VLAP Program	Begin July 2020
		B. Conduct studies to determine reservoir morphometry, bathymetric map, watershed hydrology, reservoir volume, trophic level, and flushing rate.	Claremont Water Department City of Claremont NH DES Watershed Management Bureau		
		C. Periodically Inspect the Reservoir for cyanobacteria blooms	Claremont Water Department NH DES Watershed Bureau	Claremont Water Department	Begin July 2020
		D. Conduct a build-out analysis of the Watershed to determine highest rate of possible development.	City of Claremont Upper Valley Lake Sunapee Regional Planning Commission	NH DES Regional Environmental Planning Program	2021

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
Goal #2: Reduce pollution from nonpoint sources	Objective #1: Reduce potential sources of bacteria and viruses.	A. Install coyote decoys to deter geese population at Dole Reservoir. Geese population is seasonal 40-300 geese. Determine if decoy is also needed at Whitewater Reservoir. Geese may also be harvested by Water Department staff to decrease numbers.	Claremont Water Department	Claremont Water Department	Ongoing
		B. Monitor beaver activity. Address with appropriate measures if necessary.	Claremont Conservation Commission and Water Department Skip Lisle (out of Vermont, beaver expert, "beaver deceiver")	City of Claremont	Ongoing
	Objective #2 Reduce potential sources of contamination from recreational use.	A. Develop a rotational repair schedule for bridges in the Watershed. Immediately repair the bridge approach on Skyline Drive (Class VI road) and direct water away from the tributary.	Sullivan County ATV Club/Sugah Valley Snow Riders	Department of Natural and Cultural Resources – Bureau of Trails	April 2020 In progress
		B. Implement a limited speed area within on the Watershed trail network. Re-route trails away from sensitive areas. Use trail the ATV trail cameras to assist with trail usage monitoring.	Sullivan County ATV Club/Sugah Valley Snow Riders		

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
	Objective #3 Reduce potential pollutants from forestry activities	A. Develop a new forest management plan on the Claremont Watershed properties which has preservation of water quality has the highest goal.	Three directors of Parks & Rec, Economic Dev, DPW, City Council Conservation Commission		Dec 2020
		B. On the Claremont-owned parcels, establish no harvest buffers around the reservoir and tributaries. Survey and demarcate these buffers on the ground. Coordinate buffer zones with the Town of Cornish's Wetlands Overlay district, the Shoreland Overlay district and the Shoreland Protection Act.	City of Claremont Town of Cornish Whitewater Watershed Commission Claremont Conservation Commission Cornish Conservation Commission	City of Claremont NH DES Source Water Protection grant	2021
		C. Before harvesting on the Claremont on properties, evaluate the potential impacts from forestry activities on water quality parameters such as phosphorus.	Claremont DPW Econ & Dev Parks & Rec Conservation Commission	NHDES Source Water Protection grant application deadline is Nov 1.	November 1, 2020

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
		D. Notify the Claremont Water Department when a timber harvest is scheduled to occur in the watershed.	Towns of Cornish & Croydon should notify Claremont Water Department		Ongoing
Goal #3 Ensure that the City of Claremont continues to have an adequate supply of clean, safe drinking water for the next 50 years.	Objective #1 Evaluate the adequacy of the quantity and quality of existing drinking water sources to meet future public drinking water needs.	A. Assess options for additional sources of public drinking water: For example, evaluate the feasibility of repairing the Upper Reservoir or whether to create a storage system in the Redwater Watershed.	Claremont City Council, Claremont Water Department		2021
		B. Due to water quality issues and challenges associated with treatment, phase out use of the Sugar River source. The Sugar River should be used as a source of last resort.	Claremont City Council, Claremont Water Department	N/A	2022
		C. Continue water conservation program (leak detection, water pipe cleaning) to limit water loss. Claremont is at 30% water loss.	Claremont Water Department	NH DES Leak Detection Grant	On going
		D. Implement Water Department's Energy Conservation Project	Claremont Water Department		In progress

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
	Objective #2 Conserve and protect significant and sensitive water resources in the Whitewater Watershed.	A. Identify and conserve key properties in order to protect water resources and sensitive ecological features in the watershed.	Upper Valley Land Trust Upper Valley Lake Sunapee Regional Planning Commission Watershed Conservation Commissions Claremont Water Department	NH Drinking Water Trust Fund LCHIP	Ongoing
		B. Adopt a watershed overlay district which directs development away from ecologically sensitive areas, guides construction and development and prohibits high risk land uses within the watershed boundary.	Watershed municipalities Whitewater Watershed Advisory Committee, Watershed Planning Boards, Granite State Rural Water Association	NH DES Source Water grant	2020-2025
		C. Research whether Claremont-owned parcels in the watershed permanently protected with deed restrictions	City of Claremont	NH Drinking Water Trust Fund LCHIP	2021
		D. As an ad hoc group, Whitewater Watershed Advisory Committee should carry out its collaborative efforts and work to ensure implementation of this Action Plan	Watershed municipalities, Whitewater Watershed Advisory Committee	Inkind, Watershed municipalities	Ongoing

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
Goal #4: Increase appreciation for the Whitewater Watershed and the Claremont Public Drinking Water Supply	Objective #1: Raise awareness about the Whitewater Reservoir as the primary source of Claremont's drinking water supply.	A. Present the findings of this plan to the Claremont City Council and the boards and commissions of all Watershed Municipalities.	Whitewater Watershed Advisory Committee Claremont Water Department Granite State Rural Water Association		August 2020
		B. Make paper and electronic copies of this plan available in all the Watershed communities. Distribute copies to boards and commissions, land trusts and the regional planning commission.	City of Claremont		June 2020
		C. Install a kiosk at the northern entry to the watershed that would educate all recreational users about the watershed as a drinking water source.	Claremont and Cornish Conservation Commissions Claremont Water Department Sullivan County ATV Club/Sugah Valley Snow Riders	NH DES Source Water Protection Grant	July 2021
		D. Install signage "Entering Drinking Water Watershed" at all points of entry.	Claremont Water Department	NH DES Source Water Protection Grant	July 2021
		E. The Claremont Water System will engage schools in "Drinking Water Week" celebrations.	Claremont Water Department NH DES Groundwater and Drinking Water Bureau		Annually First Week of May

Goal	Objective	Strategy	Potential Lead Agency and Partners	Potential Funding Source	Timeframe
		F. Participate in the annual NH Drinking Water Taste Test Contest at Mount Sunapee Resort.	Claremont Water Department NH DES Groundwater and Drinking Water Bureau US Department of Agriculture US Environmental Protection Agency Granite State Rural Water Association		Annually every September
		G. Send letters to Watershed property owners to inform them about the Reservoir and the need for water quality protection. Share information about the hazardous waste collection day coordinated by Upper Valley/Sunapee.	Claremont Water Department	City of Claremont	Annually starting July 2020
		H. Share the link of the video of the watershed kick off meeting on "What's up Claremont" on FaceBook, the Claremont Conservation Facebook page and Connect Cornish	Claremont Water Department CCTV		July 2020
		I. Encourage engagement and participation from the Town of Croydon in this watershed initiative	Whitewater Watershed Advisory Committee, Claremont Water Department		Ongoing

Chapter 6. Conclusion

Many different individuals, groups and agencies were involved during the process of creating this Action Plan. The next step is to share this plan with boards, citizens property owners, and other stakeholders in the Watershed municipalities. It is important that the Whitewater Watershed Advisory Committee continue to exist in order to shepherd implementation of these recommendations. As evidenced by this plan, the Committee has already had many successes and completed certain recommendations.

No planning process is complete without a periodic review of the management activities in Chapter 5. These activities should be revisited at least annually to evaluate whether strategies have been successfully implemented in order to keep the plan current and practical.

This Action Plan represents one step in a multi-stage process to safeguard the Claremont's valuable and public drinking water system. As goals are met, new ones will emerge and the Action Plan will need to be updated to reflect these changes.

Chapter 7. References

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Appendix A Cover Letter for Residential Fact Sheet Mailing

(City of Claremont Letterhead)

(Use to the Excel file to create a personal mail merge of letters)

Dear Homeowner:

The purpose of this letter is to ask for your cooperation for maintaining water quality in the Whitewater Watershed. The Watershed includes the Whitewater Reservoir, the City of Claremont's premier source of drinking water. With your support, we can ensure the quality and quantity of this source for future generations.

Your property has been identified as one of approximately 50 properties that intersect or fall completely within the Watershed. Protecting water quality is easy and makes good sense.

Enclosed are some flyers about how you can limit the use of fertilizers on your lawn, self-inspection tips for basements and heating tanks, and a homeowners guide to Septic System Maintenance. If you would like to see the full Action Plan for the watershed, please do not hesitate to call the Claremont Water Department at (603) 542-7020.

Sincerely,

Vic St. Pierre, Director
Claremont Public Works Department

Enclosures:

DES Factsheets

- Clean Drinking Water Is Up To You!
- Green Grass & Clear Water
- Self-Inspection Checklist for Basement and Outdoor Home Heating Oil Tanks
- You and Your Septic System: A Homeowner's Guide to Septic System Maintenance

Appendix B Administrative Rules

NEW HAMPSHIRE CODE OF ADMINISTRATIVE RULES

Env-Dw 902.11 Protection of the Purity of Whitewater Brook, Rice Reservoir and Dole Reservoir and Their Watersheds.

(a) The purpose of this section is to protect the purity of the water of Whitewater Brook and reservoir, Rice Reservoir and watershed and Dole Reservoir and watershed, which are principal drinking water supplies for the city of Claremont.

(b) This section shall apply within:

(1) The Whitewater Brook watershed above the Whitewater dam, which is located at approximate latitude 43° 25' 24", longitude 72° 18' 42", in the city of Claremont and towns of Cornish and Croydon;

NEW HAMPSHIRE CODE OF ADMINISTRATIVE RULES

- (2) The Rice Reservoir watershed above the Rice dam, which is located at approximate latitude 43° 25' 08", longitude 72° 19' 42", in the city of Claremont; and
 - (3) The Dole Reservoir watershed above the Dole dam, which is located at approximate latitude 43° 23' 21", longitude 72° 19' 56", in the city of Claremont.
- (c) Any person violating this section shall, in accordance with RSA 485:26, be guilty of a misdemeanor if a natural person or guilty of a felony if any other person.
- (d) In accordance with RSA 485:24, the health officers and the boards of health of the city of Claremont and the towns of Cornish and Croydon and their duly authorized agents, and the board of water commissioners of the city of Claremont and its duly authorized agents, may act as agents of the department for the enforcement of this section in cooperation with the department.
- (e) Where any provision of this section is in conflict with local ordinances, the provision that is more protective of the surface water shall apply.
- (f) Any deviations from this section shall be by written consent of the department in accordance with Env-Dw 902.05. The provisions of this section shall not apply to employees of the board of water commissioners engaged in the performance of necessary duties for the protection and control of said brook and reservoirs.
- (g) In addition to any prohibitions adopted by local ordinance, the prohibitions that apply in the Whitewater Brook, Rice Reservoir and Dole Reservoir watersheds described in (b), above shall be as follows:
- (1) A person shall not build, continue, or maintain a structure of any kind in which animals or fowl are kept within 75 feet of said brook or reservoirs or any inlet thereof or tributary thereto;
 - (2) A person shall not allow wastes or waters that have been used for washing or cleansing materials, persons, or food to run into said brook or reservoirs or any inlet thereof or tributary thereto;
 - (3) A person shall not throw or deposit any dead animal or fish or parts thereof, food, perishable or decayable material, manure, or human wastes into said brook or reservoirs or into any inlet thereof or tributary thereto, or leave or allow such materials to remain on the surface of the ground within 75 feet of said brook or reservoirs or any inlet thereof or tributary thereto;
 - (4) A person shall not throw any sawdust or allow any sawdust to fall into said brook or reservoirs or any inlet thereof or tributary thereto;
 - (5) A person shall not trespass, boat, bathe, swim, fish, or carry on any activity whatever, whether of recreational, occupational, or other nature, in the waters or on the ice of Whitewater Brook Reservoir and Intake Reservoir located on Whitewater Brook; and
 - (6) A person shall not throw, deposit, or allow to remain upon the ice of the waters of said brook or reservoirs or any inlet thereof or tributary thereto, any matter, waste or materials such as are described in (2), (3), and (4), above.
- (h) The city of Claremont shall post a summary of the prohibitions contained in (g), above, at all public access locations where persons might reasonably be expected to access Whitewater Brook, Rice Reservoir, or Dole Reservoir or their watersheds or tributaries. This posted summary may also contain any prohibitions enacted by local ordinance.

Source. (See Revision Note at part heading for Env-Dw 902)
#10602, eff 5-20-14

Appendix C List of Pertinent Statutes

The State of New Hampshire has numerous statutes and rules that are designed to protect lakes. The laws are based on the philosophy that it is “easier, cheaper, and more logical to protect lakes from degradation than it is to restore degraded lakes” (State of New Hampshire 2000 Section 305(b) Water Quality Report). Over the past two decades NH DES has made a major effort to remove point discharges of sewage and waste from lakes and from tributaries to lakes. A brief summary of some of the laws and regulations that help protect New Hampshire lakes is presented in Table 3.1

Table 3.1 List of pertinent statutes established to protect New Hampshire lakes. (Source: State of New Hampshire 2000 Section 305(b) Water Quality Report).

Provision	Regulatory Authority
1. All lakes are classified at least Class B (RSA 485-A:11), the goal is that these waterbodies are suitable for fishing, swimming, and other recreational activities (RSA 485-A:8-II), and violations of assigned classifications are not allowed (RSA 485-A:12-II).	RSA 485-A:11 RSA 485-A:8-II RSA 485-A:12-II
2. No discharge is allowed to a lake without a permit. It is prohibited to discharge marine toilets into a lake (RSA 487:2).	RSA 485-A:13-I
3. Dumping of trash in or on the banks of a lake is prohibited.	RSA 487:2
4. Graywater (sink and shower wastes) from boats cannot be discharged into a lake.	RSA 487:3
5. No new point sources of phosphorus to lakes are allowed, and no new discharges of phosphorus to tributaries of lakes are allowed that would encourage weed or algae growth.	WS432.10
6. Existing high quality lakes shall be maintained at their existing high quality.	WS439.02

Provision	Regulatory Authority
7. No automobiles may be washed in or driven into any lake.	Uncodified regulation – may not be enforceable.
8. Automobiles and other petroleum powered vehicles lost through the ice into a lake must be removed.	RSA 485-A:14
9. Dredge and fill activities are not allowed in or around a lake without a permit.	RSA 482-A:3, 485-A:17
10. No construction or transportation of forest products (skidding etc.) can occur near a lake without a permit.	RSA 485-A:17
11. No earth moving activities are allowed near a lake without a permit.	RSA 485-A:17
12. No subsurface disposal system may be installed near a lake without a permit and without meeting minimum standards.	RSA 485-A:29
13. No pesticides applications are permitted within 25 feet of lakes without a permit (RSA 430:28-48) and the recommendation of DES (Pes 502, 601, 604).	RSA 430:28-48 Pes 502, 601, 604
14. Cottages near lakes or tributaries to lakes cannot be converted from seasonal to year-round use unless an application for approval of the sewage disposal system has been submitted and approved.	RSA 485-A:38
15. Cottages near lakes or tributaries to lakes cannot be expanded in size such that the load on the sewage disposal system is increased, unless an application for approval of the sewage disposal system is submitted.	RSA 485-A:38
16. No property with a sewage disposal system located within 200 feet of a great pond can be offered for sale until a licensed sewage disposal designer has performed a site assessment to determine if the site meets current standards for sewage disposal systems.	Relevant Law: RSA 4:40-a, 485-A:2, RSA 485-A:39 Relevant Adm. Rule: Env-Ws 1025
17. The Lakes Management and Protection Program established a lakes coordinator and lakes management advisory committee to prepare: (1) statewide lake management criteria and (2) guidelines for the development of local lake management and shoreland protection plans.	RSA 483-A
18. The Shoreland Protection Act provides minimum protective standards for activities occurring within 250 feet of lakes and ponds with a surface area of 10 acres or more.	RSA 483-B
19. No household cleaning products except those used in dishwashers shall be distributed, sold, or offered for sale in New Hampshire which contains a phosphorus compound in excess of a trace quantity.	RSA 485-A:56
20. No exotic aquatic weeds shall be offered for sale, distributed, sold, or imported, purchased, propagated, transported, or introduced in the state.	RSA 487:16a
21. Permits are also required for the following activities, and permits will not be issued if lake quality is to be endangered: a. Groundwater discharges b. Underground storage tanks c. Solid waste landfills d. Sludge pits e. Hazardous waste sites	RSA 485-A:13 RSA 146-A RSA 149-M RSA 149-M RSA 147-A

Appendix D. List of Property Owners in the Whitewater Watershed.

Town	Map	Lot	Sub	Owner	Co-Owner	Address	City	State	Zipcode	Acreage
Claremont	18	9		PAIGE RICHARD E	PAIGE SARAH	716 FURMAN AVE	CORPUS CHRISTI	TX		74.9
Claremont	18	7		PAIGE RICHARD E	PAIGE SARAH	716 FURMAN AVE	CORPUS CHRISTI	TX		33.6
Claremont	54	4		IANNUCCI TRUST FRANK H JR	IANNUCCI JR FRANK H TRUS*	99 HILLSIDE DR	PROVIDENCE	RI		252.6
Claremont	20	2		BLUE MOUNTAIN FOREST ASSOC		154 SOUTH RD	NEWBURY	NH		39.8
Claremont	31	3		LESKIEWICZ JR EDWARD H		73 THRASHER RD	CLAREMONT	NH		6.1
Claremont	31	2		LESKIEWICZ JR EDWARD H		73 THRASHER RD	CLAREMONT	NH		19.8
Claremont	32	1		ARGOE MARTHA LEWIN		100 FACTORY CREEK COURT	BEAUFORT	SC		22.9
Claremont	31	4		BLUE MOUNTAIN FOREST ASSOC		154 SOUTH RD	NEWBURY	NH		21.6
Claremont	20	1		BROWN PK AND SON INC		PO BOX 923	CLAREMONT	NH		57.2
Claremont	42	1		CELANI JONATHAN		20 BLUEBERRY LN	HUDSON	NH		296.6
Claremont	19	2		DIERKS ANGELICA C		874 MORNINGSIDES LN	CHARLESTOWN	NH		94.5
Claremont	19	1		DIERKS ANGELICA C		874 MORNINGSIDES LN	CHARLESTOWN	NH		36.7
Claremont	18	2		CITY OF CLAREMONT		CITY HALL	CLAREMONT	NH		30.7
Claremont	8	2		CITY OF CLAREMONT		CITY HALL	CLAREMONT	NH		3.6
Claremont	8	1		CITY OF CLAREMONT		CITY HALL	CLAREMONT	NH		10.0
Claremont	7	7		THEDFORD HENRY A	THEDFORD JAYNE M	91 WHITEWATER BROOK RD	CLAREMONT	NH		15.4
Cornish	11	38	0	RIVERBROOK INVESTMENTS, LLC		166 SKYLINE DRIVE	CORNISH	NH	03745	4.1
Cornish	11	39	0	MEYETTE LIVING TRUST	JOSEPH & PEGGY	11 JACKSON RD	CORNISH	NH	03745	179.6
Cornish	11	41	0	MEYETTE TRUSTEE, BRIAN H	MEYETTE TREE FARM	166 SKYLINE DR	CORNISH	NH	03745	25.4
Cornish	11	40	0	CHICOINE, DEVRA	LABRECQUE, KATHRYN	76 SKYLINE DRIVE	CORNISH	NH	03745	51.1
Cornish	11	42	0	MEYETTE, BRIAN H.		166 SKYLINE DR	CORNISH	NH	03745	40.5
Cornish	11	41	0	MEYETTE TRUSTEE, BRIAN H	MEYETTE TREE FARM	166 SKYLINE DR	CORNISH	NH	03745	25.4
Cornish	11	47	0	NEIL REV. TRUST, JAMES H.	JAMES H. NEIL, TRUSTEE	813 EAST RD	CORNISH	NH	03745	75.9
Cornish	11	49	0	MEYETTE LIVING TRUST	JOSEPH & PEGGY	11 JACKSON RD	CORNISH	NH	03745	0.2

Cornish	12	27	0	RUSHIA, LEO G.		PO BOX 46	NORTH HARTLAND	VT	05052	75.3
Cornish	12	28	0	SICARD, BRIAN P	SICARD, DEBORAH L	1362 NH RTE 120	CORNISH	NH	03745	27.0
Cornish	12	29	A	BERGERON, MARK J	BERGERON, TAMMY L	21 MELLOW MT RD	CORNISH	NH	03745	19.7
Cornish	12	29	C	BRODEUR, JANE		17 MELLOW MT RD	CORNISH	NH	03745	7.4
Cornish	12	29	D	FORMICHELLI, JOHN JAMES		28 MELLOW MT. RD	CORNISH	NH	03745	13.7
Cornish	12	29	B	LOWE, JOHN		16 MELLOW MT ROAD	CORNISH	NH	03745	10.7
Cornish	12	29	0	SIMINO, TROY	SIMINO, ALICIA	PO BOX 228	CORNISH FLAT	NH	03746	17.6
Cornish	12	31	0	HAWKINS REV TRUST	HAWKINS, RALPH & DIANE CO-TRSTEES	32 SUNNY LANE	SUNAPEE	NH	03782	59.2
Cornish	12	32	A	BONNEAU, HEATHER M	& BRADLEY S	1181 NH ROUTE 120	CORNISH	NH	03745	5.5
Cornish	12	29	E	SIMINO, TROY	SIMINO, ALICIA	PO BOX 228	CORNISH FLAT	NH	03746	17.3
Cornish	12	30	0	SICARD, BRIAN P	SICARD, DEBORAH L	1362 NH RTE 120	CORNISH	NH	03745	21.8
Cornish	12	32	0	LUKASH, JAMES R.	LUKASH, PAMELA J.	13 HUNT DR	CORNISH	NH	03745	17.9
Cornish	12	33	0	GOBIN, WALTER C.	GOBIN, SANDRA L.	1161 NH RTE 120	CORNISH	NH	03745	8.9
Cornish	12	35	0	MEYETTE TRUSTEE, JOSEPH C	MEYETTE TRUSTEE, MARGARET E	11 JACKSON RD	CORNISH	NH	03745	54.9
Cornish	12	36	0	BEAN, MARTIN H	WRIGHT, GRETA C	PO BOX 404	WILMOT	NH	03287	35.6
Cornish	12	37	0	FINE, PATRICK C.	FINE, SUSAN F.	500 N. DUKE STREET #53-102	DURHAM	NC	27701	35.9
Cornish	12	37	A	OLIVIER, BRIAN C	FIFIELD, J. PATRICK	3A HERITAGE WAY	RAYMOND	NH	03077	31.9
Cornish	12	38	0	CLAREMONT, CITY OF	PUBLIC WORKS DEPARTMENT	WATER DIVISION	CLAREMONT	NH	03743	86.5
Cornish	12	39	0	CORNISH, TOWN OF		488 TOWN HOUSE RD	CORNISH	NH	03745	5.2
Cornish	12	40		RENE GAFFNEY	60 KENDALL STREET		FRANKLIN	NH		17.1
Cornish	12	41	A	BARTH, CONSTANCE & JOHN	ECKRICH, SARAH	1858 BIBLE HILL ROAD	BROWNSVILLE	VT	05037	20.3
Cornish	12	41	0	BARTH, CONSTANCE & JOHN	ECKRICH, SARAH	1858 BIBLE HILL ROAD	BROWNSVILLE	VT	05037	7.0
Cornish	12	43	0	CLAREMONT, CITY OF	PUBLIC WORKS DEPARTMENT	WATER DIVISION	CLAREMONT	NH	03743	320.9
Cornish	12	44	0	CORNISH, TOWN OF		488 TOWN HOUSE RD	CORNISH	NH	03745	67.9
Cornish	13	1		BLUE MOUNTAIN FOREST ASSOC		PO BOX 487	NEWPORT	NH	03773	14.6
Cornish	14	1	0	BLUE MOUNTAIN FOREST ASSOC		PO BOX 487	NEWPORT	NH	03773	1858.2
Cornish	14	1	0	BLUE MOUNTAIN FOREST ASSOC		PO BOX 487	NEWPORT	NH	03773	17.0
Croydon				BLUE MOUNTAIN FOREST ASSOC		PO BOX 487	NEWPORT	NH	03773	721.0

